

# Introduction to Geometrical Optics

## I. Fundamental Laws of Geometrical Optics

### I.1 Definition

Optics is a branch of physics that studies visible light, its behavior, its propagation, and its various properties. It is concerned both with phenomena related to the emission and use of light and with those of human vision.

This discipline analyzes the laws governing luminous phenomena, particularly those perceived by the eye. It makes it possible to understand how the information transmitted to the eye such as the shape, color, or position of an object is interpreted by our visual system.

Among the different approaches to optics, geometrical optics focuses on the path of light, taking into account the properties of the media it travels through. It makes it possible, for example, to explain phenomena such as reflection or refraction using simplified models in which light is represented by rays.

### I.2 Nature of Light

Light carries energy in the form of an electromagnetic wave in a vacuum or in a transparent medium. It generally results from the superposition of waves with different wavelengths. Monochromatic light corresponds to a single sinusoidal wave with a well-defined frequency, whereas polychromatic light is composed of several electromagnetic waves.

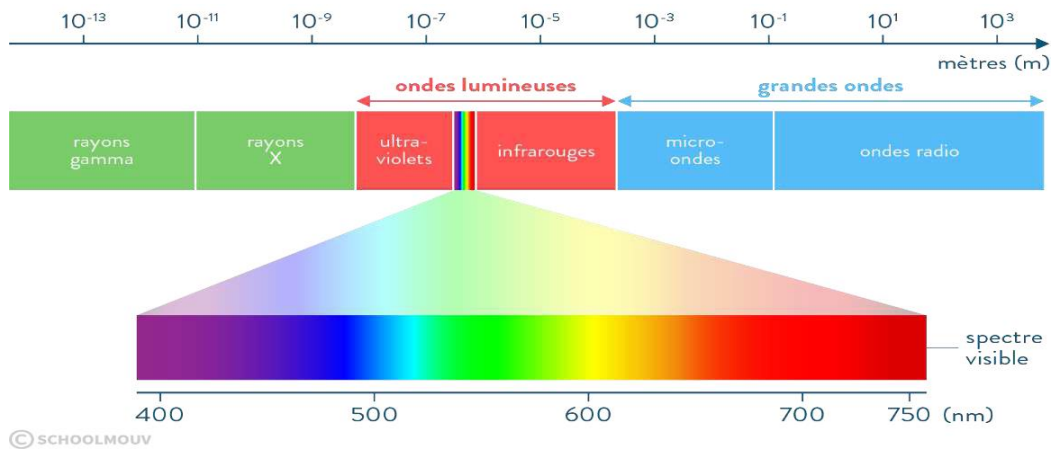
- In a vacuum, light propagates in all directions of space at the speed:

$$c = 299792458 \text{ m/s} \approx 3 \cdot 10^8 \text{ m/s}$$

- The wavelength  $\lambda$  in a vacuum, the frequency  $\nu$ , and the period  $T$  are related by:

$$\lambda = c \cdot T = c/\nu .$$

- The range of light visible to the human eye corresponds to wavelengths between  $0.4 \mu\text{m}$  and  $0.8 \mu\text{m}$  (400 nm and 800 nm).



**Fig. I.1. Spectra of light.**

### **I.3. Principle of rectilinear propagation of light**

#### **I.3.1 Light sources**

A light source is an object or a device that produces light. There are two main types:

##### **I.3.1.1 Primary light source**

It is an object that produces its own light; it emits light directly without needing to be illuminated by another source. Examples are the Sun, a candle, and similar sources.

##### **I.3.1.2 Secondary light source**

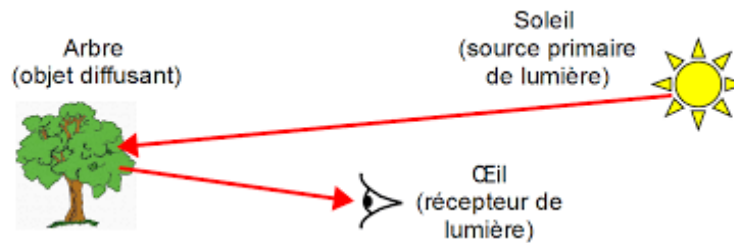
A secondary source (also called a diffusing object) is an object that does not produce its own light. It becomes visible only when it is illuminated by a primary source, as it reflects or diffuses the light it receives.

Diffusion is a phenomenon in which an object receives light and then reflects or scatters part or all of that light in all directions. A diffusing object does not emit its own light. It becomes visible only when illuminated by a primary source (such as the Sun or a lamp) or by another diffusing object.

For example:

- The Moon is a diffusing object: it does not produce light but reflects the Sun's light.

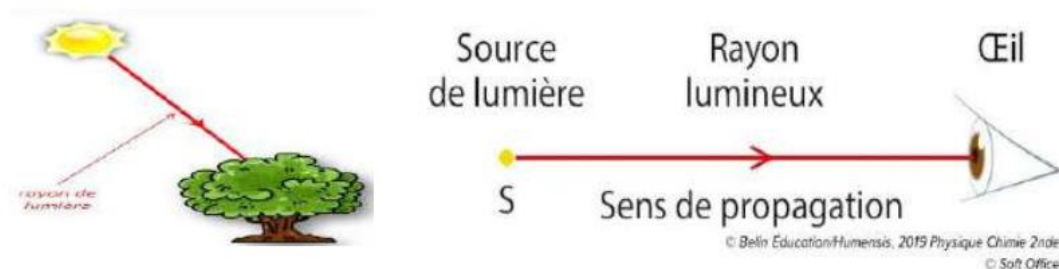
- The planets in our solar system are also diffusing objects.
- All objects around us, including people, walls, books, etc., are visible because they diffuse the light they receive.



**Fig. I.2. Theory of light ray vision**

### I.3.2. Light ray

Light is described as a collection of independent light rays. These light rays are characterized by a direction of propagation and a propagation speed ( $v$ ). The rays (emitted from a source) travel in straight lines through any homogeneous medium at a speed that depends on the medium.



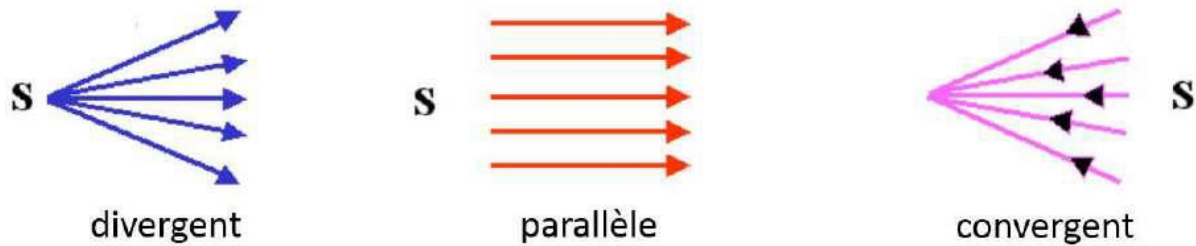
**Fig. I.3. 3. Light ray**

A light beam is a collection of light rays that propagate in a given direction.

It can be:

- **Convergent:** the rays move toward the same point (they come closer together).

- **Divergent:** the rays move away from each other, starting from a common point.
- **Parallel:** the rays remain parallel to each other, neither converging nor diverging.



**Fig. I.4. Light beam**

#### Examples:

- The light from a flashlight forms a divergent beam.
- The light emitted by a laser is an almost perfectly collimated parallel beam.
- A concave mirror can converge light rays to a focal point.

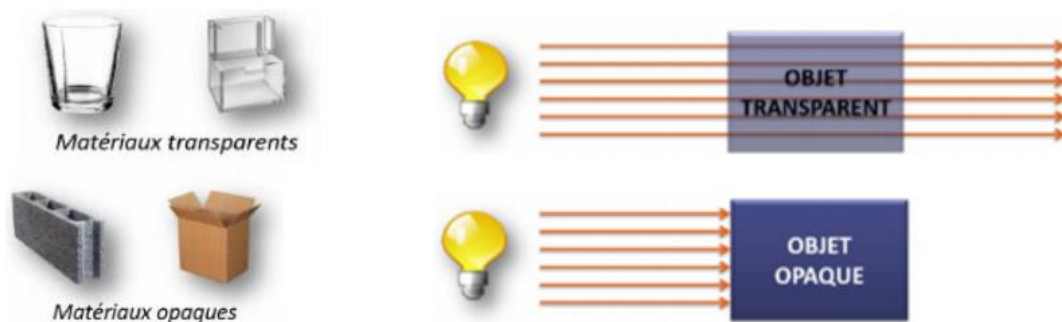
**Note:** Light propagates through a transparent medium.

#### I.4. Types of media

**a. Transparent media:** Objects are seen clearly: air, water, glass.

**b. Opaque media:** Objects cannot be seen: wall, wood, cardboard.

**c. Translucent media:** Allows light to pass but objects are not seen clearly: frosted glass, tracing paper.



**Fig. I.5. Media for light propagation**

## I.5. Refractive index of a medium

The propagation speed (or velocity) of a wave depends on the medium it passes through. In a medium other than vacuum, it is less than (c):

$$v = \frac{c}{n} \quad \text{with } n > 1.$$

The quantity (n) is called the **refractive index** of the medium.

- (n) = refractive index
- (c) = speed of light in vacuum ( $\approx 3 \cdot 10^8$  m/s)
- (v) = speed of light in the medium

### Note:

- The larger n is, the more the light is slowed down in that medium.
- The refractive index n is always  $\geq 1$  (because light is always fastest in a vacuum compared to any other material).
- The refractive index n depends on the nature of the transparent media in contact and on the color (wavelength) of the light.

### Examples of refractive indices:

- **Air:** ( $n \approx 1.0003$ )
- **Water:** ( $n \approx 1.33$ )
- **Glass:** ( $n \approx 1.5$ )
- **Diamond:** ( $n \approx 2.42$ )

A medium whose refractive index is higher than that of another medium is said to be **more refractive**.

## I.6. Optical path (Fermat's Principle)

Fermat's principle (also called the **principle of least time**) is a concept in optics, related to light. It states simply:

Light always follows the path that takes the **least time** to travel from one point to another.

This is not necessarily the shortest path in distance, but the **fastest path**.

In medicine, this principle helps to understand the behavior of light in biological tissues, lenses (such as in the eye or medical instruments), and also in medical optical examinations (ophthalmology, imaging, etc.).

### **Medical application**

#### **Example in the human eye:**

Light enters the eye, which is composed of several transparent media, passing through the cornea, aqueous humor, lens, etc. Each medium has a different refractive index (and therefore a different speed for light). The light changes direction at each interface to **minimize its travel time** to the retina. This is a direct application of **Fermat's principle**.

#### **Concrete example:**

- If an object is in front of us, the light coming from that object enters the eye through the cornea.
- Due to differences in refractive index (air  $\approx 1$ , cornea  $\approx 1.376$ ), the light is strongly refracted at the cornea.
- It continues to change direction in the other media so that all rays converge on the same point of the retina. (The light adjusts its path to reach the correct spot on the retina in minimal time.)
- The brain interprets this convergence as a sharp image.

#### **Note :**

In a homogeneous medium, Fermat's principle simply becomes: **“Light travels in a straight line.”** This is a special case of the principle.

### **I.7. Reflection and refraction phenomena: laws of reflection and refraction**

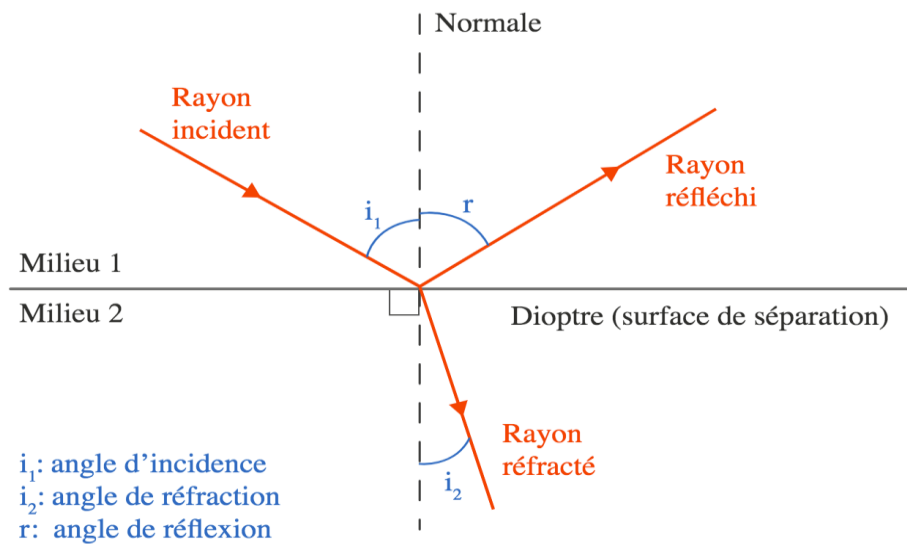
A **dioptr** is the surface separating two transparent media with different refractive indices. Light rays remain straight in a homogeneous and isotropic medium, but they are **deviated** when crossing a dioptr or encountering a reflective surface.

When an incident beam reaches the dioptr at the point of incidence, a **reflected beam** and a **refracted beam** may appear.

The plane containing the incident beam, the reflected beam, and the refracted beam is called the **plane of incidence**. It also contains the **normal** to the dioptr at the point of incidence.

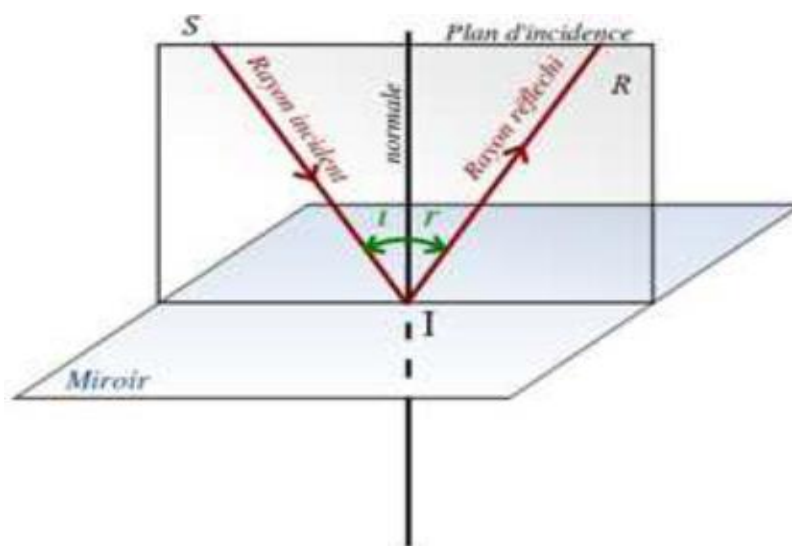
- **Reflection** involves a change in the direction of the ray at a boundary surface, but **without changing the medium** (the incident ray and the reflected ray travel in the same medium).

- **Refraction** refers to the deviation of a ray when it **crosses the boundary between two media** (the incident ray and the refracted ray travel through different media).



**Fig.I.6.**

- **Point of incidence:** the point where the incident ray strikes.
- **Normal:** a line passing through the point of incidence and perpendicular to the tangent plane at that point.
- **Angle of incidence:** the angle between the normal and the incident ray.
- If the incident light is mostly **refracted** → **dioptr** (separating surface)
- If the incident light is mostly **reflected** → **mirror** (reflective surface)



**Fig.I.7.**

## II. Snell-Descartes Laws

They specify the directions of the reflected and refracted rays at the interface between two transparent media with refractive indices  $n_1$  and  $n_2$ .

### 1. Laws of Reflection

#### First law of Descartes:

The incident ray, the reflected ray, and the normal to the interface lie in the plane of incidence.

#### Second law of Descartes:

The angle of reflection  $i'$  is equal in magnitude to the angle of incidence  $i$ .

$$i = i'$$

These angles are measured relative to the normal, not relative to the surface.

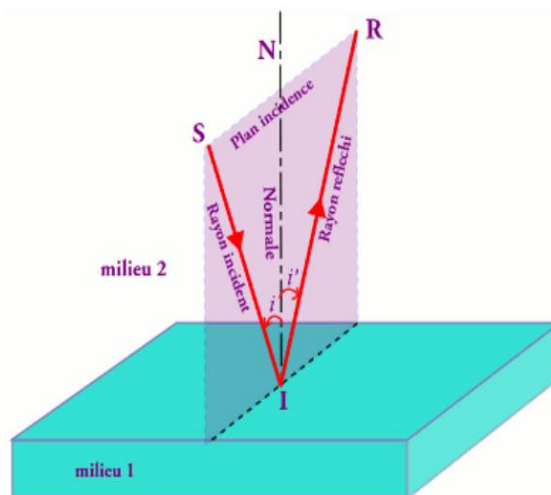


Fig. II.1. Reflection of a light ray at an interface.

### 2. Laws of Refraction:

#### First law of refraction:

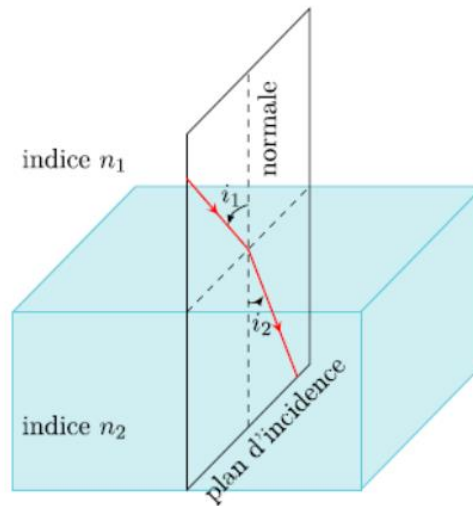
The incident ray, the refracted ray, and the normal to the interface lie in the plane of incidence.



**Second law of refraction:**

The angle of incidence and the angle of refraction are related by the second Snell–Descartes law:

$$n_1 \sin i_1 = n_2 \sin i_2$$



**Fig. II.2. Refraction of a light ray at an interface.**

**3. Kepler's Law**

In the case where the light ray is close to the normal, the angles of incidence and refraction become small, and the sine of the angle can be approximated by the angle itself. One thus obtains Kepler's law:

$$n_1 i_1 = n_2 i_2$$

**4. Critical Refraction. Total Internal Reflection**

The laws stated above determine only the direction of the refracted ray, given that of the incident ray. We will now revisit the study of refraction in order to specify certain aspects, taking into account the existence of reflected light.

Light passes from a medium with refractive index  $n_1$  to another medium with refractive index  $n_2$ .

Case:

### $n_1 < n_2$ : limiting refraction

The medium with refractive index  $n_2$  is more optically dense than the medium with refractive index  $n_1$ .

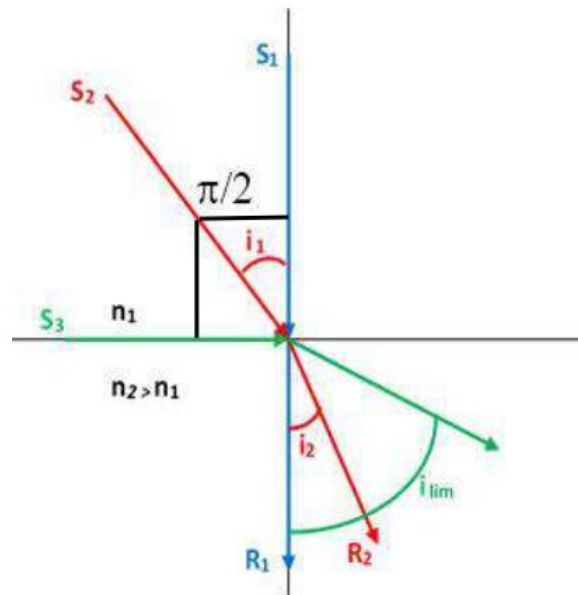
$$\frac{n_1}{n_2} < 1 \Rightarrow \frac{\sin(i_2)}{\sin(i_1)} < 1 \Rightarrow \sin(i_2) < \sin(i_1) \Rightarrow i_2 < i_1$$

If the angle of incidence  $i_1$  is gradually increased, it is observed that the angle of refraction increases with the angle of incidence, but more slowly, so that  $i_2 < i_1$ . The refracted ray approaches the normal.

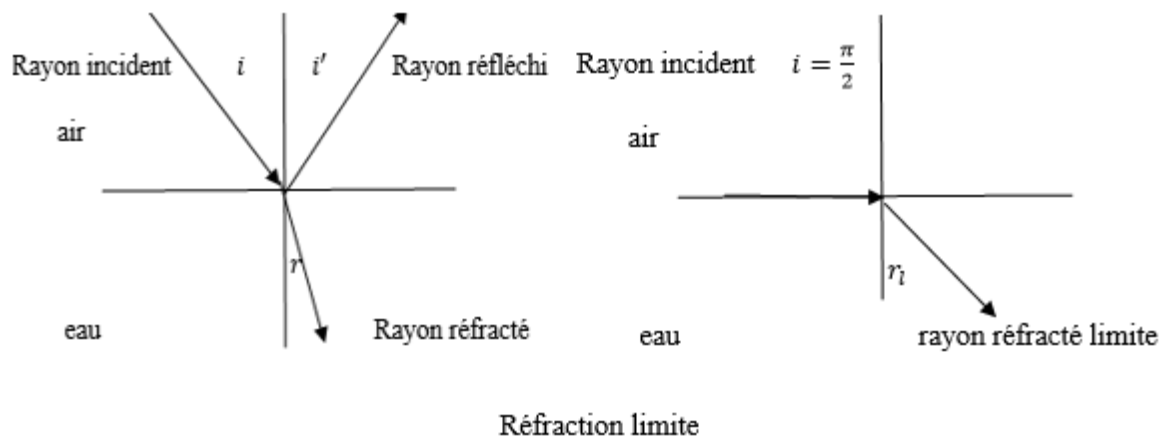
If  $i_1 = \pi/2$  (the angle of incidence is maximum), then a limiting angle of refraction  $i_{lim}$  is obtained, which can be calculated as:

$$n_1 \sin \pi/2 = n_2 \sin i_{lim}$$

$$\sin i_{lim} = \frac{n_1}{n_2}$$



**Fig. II.3. Refraction when  $n_1 < n_2$ .**



**Fig.II.4.**

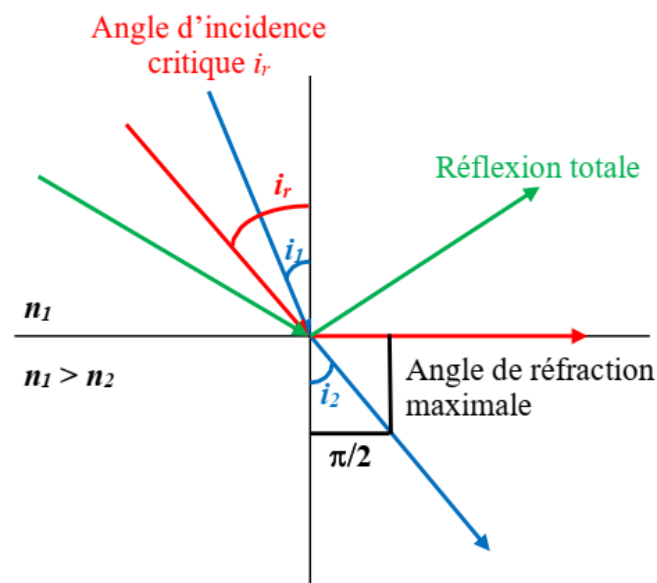
**Case:**

**$n_1 > n_2$ : total internal reflection**

The medium with refractive index  $n_1$  is more optically dense than that with refractive index  $n_2$ .

$$\frac{n_1}{n_2} > 1 \Rightarrow \frac{\sin(i_2)}{\sin(i_1)} > 1 \quad \sin(i_2) > \sin(i_1) \Rightarrow i_2 > i_1$$

As the angle of incidence  $i_1$  is gradually increased, the angle of refraction also increases, and the refracted ray moves further away from the normal. At a certain value of  $i_1$ , the angle of refraction reaches its maximum value:  $i_2 = \pi/2$ .



**Fig.II.5**

When :

$$i_2 = \pi/2 \rightarrow i_1 = i_r$$

$$n_1 \sin(i_r) = n_2 \sin(\pi/2)$$
$$\sin(i_r) = \frac{n_2}{n_1}$$

The angle  $i_r$  is called the critical angle of incidence. When  $i < i_r$ , both refraction and reflection take place. When  $i > i_r$ , only reflection occurs; this is known as total internal reflection

When the angle of incidence exceeds  $i_r$ , no refracted ray is produced; the incident ray is entirely reflected. This phenomenon is called total internal reflection. In this case, the interface behaves like a mirror.

**Note:**

**Whenever refraction occurs, reflection always accompanies it. However, reflection does not always involve refraction.**

## 5.APPLICATIONS:

### OPTICAL FIBER:

An optical fiber is a very thin strand of glass or plastic capable of transmitting light (or light signals) over long distances with minimal loss. The light is guided inside the fiber through a phenomenon called **total internal reflection**. Here's how it works:

- **Structure of the fiber:**
  - **Core:** the central part, with refractive index  $n_1$
  - **Cladding:** the outer layer, with refractive index  $n_2$
  - It is always true that  $n_1 > n_2$ , ensuring total internal reflection inside the core.
- **Propagation of light:**

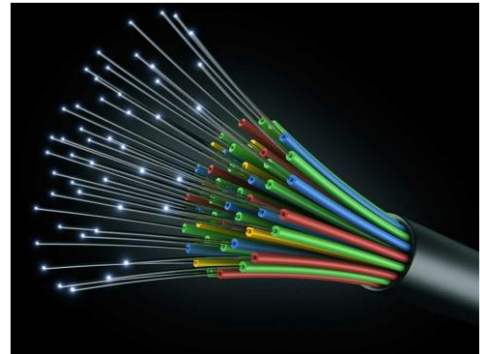
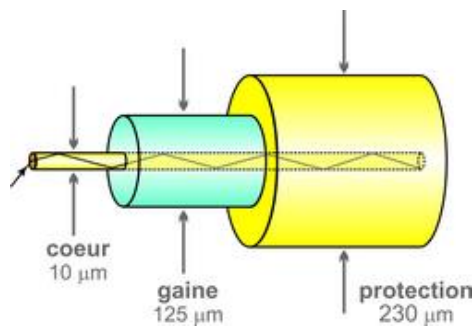
When a light ray enters the fiber at one end with a suitable angle, it undergoes multiple total internal reflections. The ray then propagates to the other end of the fiber without loss, following a zigzag path.
- **Light characteristics:**

Light is an electromagnetic wave. The visible range of wavelengths is from 390 to 760 nm. Wavelengths above 760 nm are **infrared**, and below

390 nm are **ultraviolet**. The three wavelengths most commonly used in optical fibers are 850, 1310, and 1550 nm.

- **Light source:**

The light used is generally an **infrared laser** (with a specific wavelength).



**Fig.II.6.**

## **Fibroscope**

Fibroscope allows the physician to:

- **Observe the interior of an organ in real time.**

A **fibroscope** is a flexible (or rigid) tube containing multiple optical fibers.

When a light ray enters the core of a fiber at a sufficiently large angle, it undergoes **total internal reflection** each time it strikes the wall:

- It is reflected inside the core
- It continues to propagate in a zigzag path without escaping
- Even if the fiber is bent, the light remains trapped inside

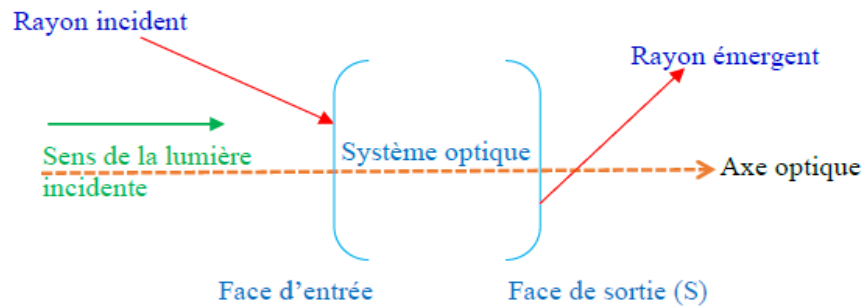
The fibroscope is equipped with an optical system (camera) and a channel that allows for possible sample collection.



### III. Stigmatism, Aplanatism, Diopters, and Mirrors

#### 1. Optical Systems

In geometrical optics, an optical system is any element capable of altering the path of light.



**Fig. III.1. Representation of an Optical System.**

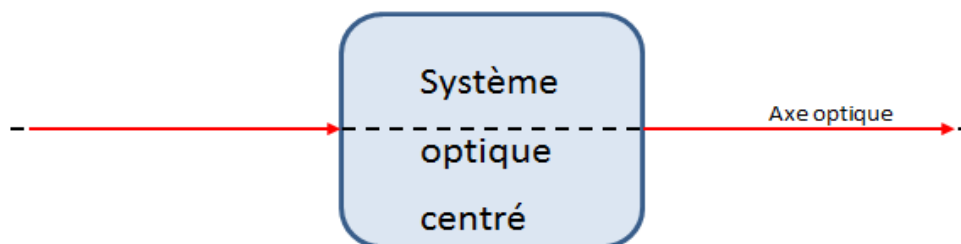
An optical system transforms an incident light ray into an emerging ray in a direction that is, in general, different from the incident direction.

An optical system can take one of the following forms:

- Dioptric system: an optical system consisting only of diopters (lenses, eyeglasses, microscopes).
- Catadioptric system: an optical system composed of both diopters and mirrors (telescopes).
- Catoptric system: a system consisting only of mirrors.

#### 1.1. Centered Optical System and Optical Axis:

A **centered optical system** is an optical system that has a symmetry axis, called the **optical axis**. A centered optical system can be composed of one or more diopters and/or mirrors.



**Fig.III.2.**

- Rays incident along the optical axis are not deviated by the optical system.

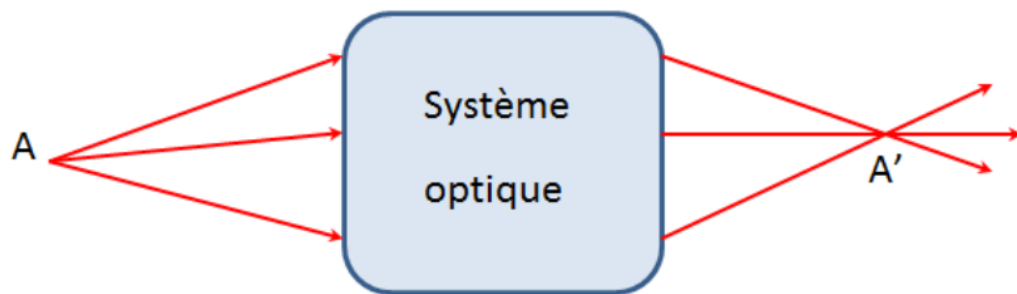
## 2.OBJECT:

For a given optical system, an object point is the point of intersection of the incident rays.

## 1.3.IMAGE:

For a given optical system, an image point is the point of intersection of the emergent rays.

Let A be a light source emitting rays toward an optical system (S). If all rays originating from A converge at a point A' after passing through (S), then A' is said to be the image of A with respect to (S). By application of the principle of reversibility of light, A and A' are said to be conjugate through (S).



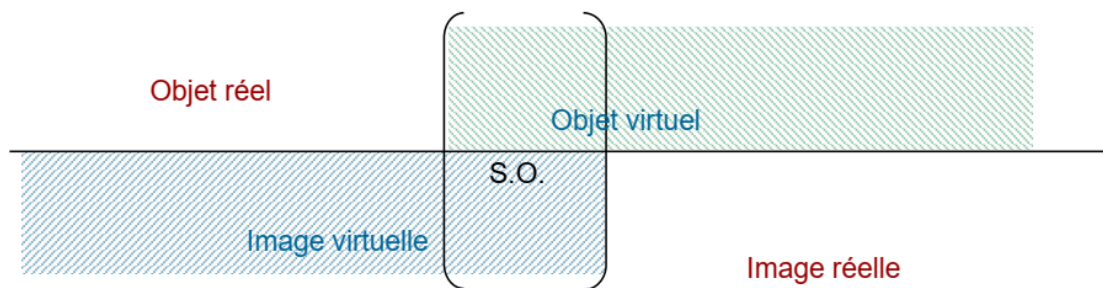
**Fig. III.3. Example of an object point A and an image point A'**

- No optical system  $\rightarrow$  no object, no image.

## 1.4. Object space and image space

An object is said to be real if it is located in the object space of the optical system, that is, in front of the entrance surface of the system. Otherwise, it is said to be virtual.

An image is said to be real if it is located in the image space of the optical system, that is, beyond the exit surface of the system. Otherwise, it is said to be virtual.



**Fig. III.4. Object space and image space.**

**Note:** A virtual image **cannot** be observed directly on a screen. If we consider only a single optical system, a real image **can** be observed on a screen.

To clarify:

- **Real image:** This is the image formed when **converging light rays** actually meet at a point after passing through an optical system (like a converging lens or concave mirror).
  - It can be **projected onto a screen**.
  - Example: the image of an object projected onto a screen using a converging lens.
- **Virtual image:** This is the image formed when the light rays **appear to diverge** from a point but **never actually meet**.
  - It **cannot be projected onto a screen**.
  - Example: the image of an object seen through a magnifying glass or a plane mirror.

💡 Memory tip:

- *Real → Really on a screen*
- *Virtual → Only visible by looking into the optical system*

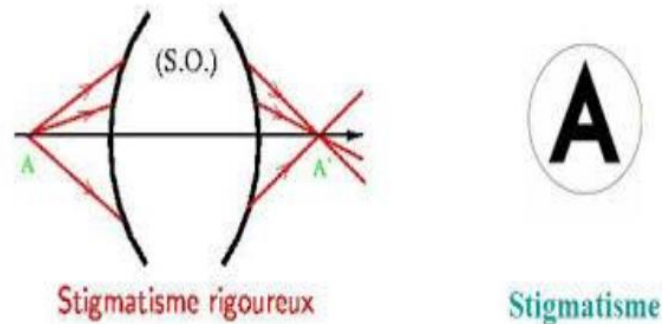
## 2. Stigmatism

An optical system is of good quality if it produces a point image from a point source: this is the condition of stigmatism.

### 2.1. Strict Stigmatism



An optical system is strictly stigmatic (Figure II.1) if all the rays emitted from the object point (A) converge to a single image point (A'). In other words, when it produces a sharp, high-quality image, we say that A and A' are conjugate points of the optical system.



**Fig. III.5. Stigmatism**

## 2.2. Approximate Stigmatism

Since cases of strict stigmatism are rare, perfect optical instruments are unrealizable. In practice, we must settle for approximate stigmatism, which is an idealization.

An optical system (S) exhibits approximate stigmatism for a pair of points (A; A') if every incident ray passing through A emerges, after passing through S, in the vicinity of A'. In this case, the image of A is a small spot centered on A', and we say that A and A' are conjugate in the sense of approximate stigmatism.

Most optical systems produce relatively good-quality images, achieved by adjustments and apertures. This is what is called approximate stigmatism.

### Gauss Approximation:

A centered optical system will produce a good-quality image of an object if the following two conditions, called Gauss conditions, are satisfied:

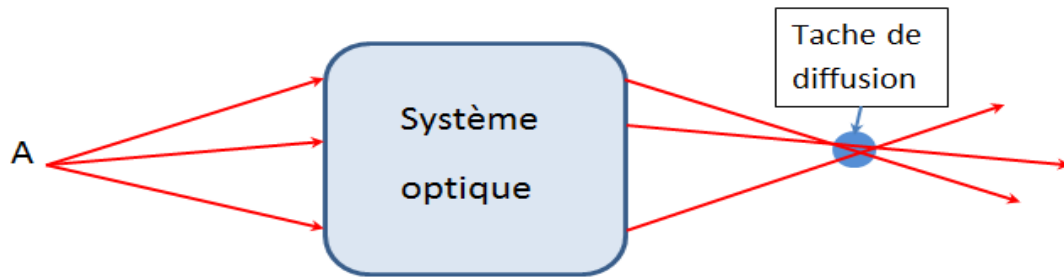
**Condition 1:** The incident rays are very close to the optical axis.

**Condition 2:** The incident rays are only slightly inclined with respect to the optical axis.

$$n_1 \cdot \sin(i_1) = n_2 \cdot \sin(i_2) \longrightarrow n_1 \cdot i_1 = n_2 \cdot i_2$$

The Gauss conditions ensure that centered optical systems exhibit approximate stigmatism (point-to-point conjugation) and approximate aplanatism (plane-to-plane conjugation).

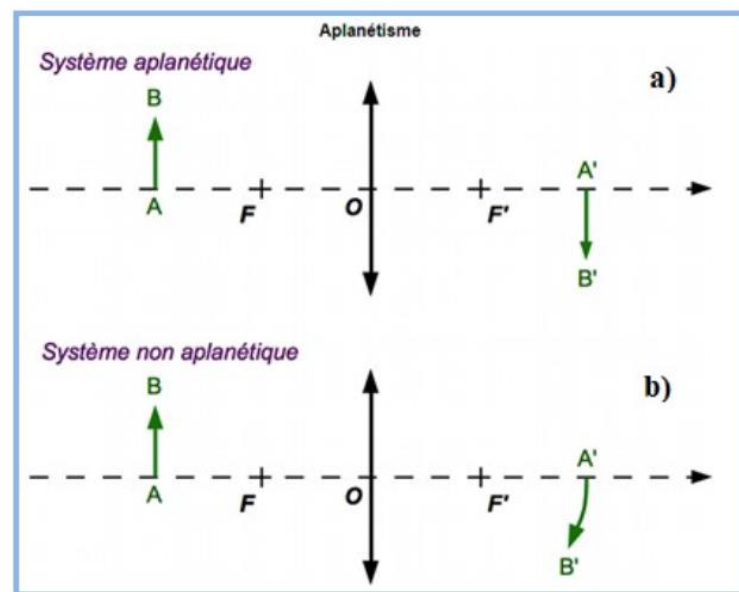
Stigmatism allows us to associate a point on the axis with an image on the axis: a characteristic conjugation relationship expresses this property.



**Fig. III.6. Approximate Stigmatism**

### 2.3. Aplanatism

Aplanatism is a property of dioptric, catoptric, and catadioptric optical systems that allows them, for an extended object perpendicular to the optical axis, to form an image perpendicular to the optical axis.



**Fig. III.7. System a) aplanatic and b) non-aplanatic.**

The object AB is perpendicular to the optical axis. In the upper case, the image A'B' is also perpendicular to the axis; the system is aplanatic. In the lower case, it is not perpendicular to the axis; it is distorted. The system is not aplanatic.

### **3. Properties of centered systems**

#### **3.1. Conjugation relation:**

The system produces, from an object point A on the axis, an image A' also on the axis. The position of A' depends on that of A. Therefore, there is a mathematical relationship linking the positions of A and A'. This relationship is called the “conjugation relation.”

#### **3.2 Magnification**

The transverse linear magnification  $\gamma$  is defined as the ratio of the algebraic values of the linear dimensions of the image A'B' to those of the object AB:

$$\gamma = \frac{\overline{A'B'}}{\overline{AB}}$$

$\gamma$  is a dimensionless algebraic value, positive if the image and the object have the same orientation, and negative if the image is inverted relative to the object.

## IV. Plane Diopter

### I. Definition

A diopter is a surface separating two homogeneous and isotropic transparent media with different refractive indices.

Diopters are refractive elements.

A diopter is called **plane** if the separating surface is a plane, and **spherical** if it is a sphere (or at least a spherical cap).

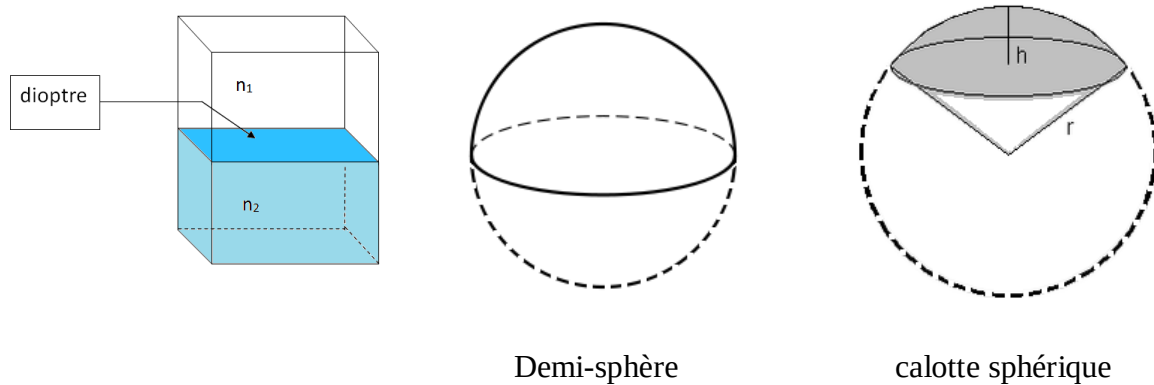


Fig.IV.1.

Thus, a plane diopter is a flat surface separating two transparent, homogeneous media with different refractive indices, which forms, by refraction, the image of a given object.

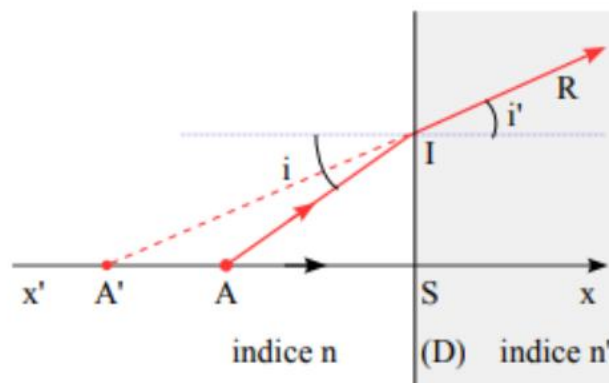


Fig.IV.2.

Consider an object point A in a medium with refractive index  $n$ . Since the system is rotationally symmetric around the normal AS, the ray AS passes through the surface without deviation. If an image A' exists, it must lie on AS. Let an incident ray AI strike the diopter with an angle of incidence  $i$ . The corresponding refracted ray IR intersects AS at A' such that:

$$SI = SA \cdot \tan(i) = SA' \cdot \tan(i')$$

$$SA' = SA \cdot \frac{\sin(i)}{\cos(i)} \cdot \frac{\cos(i')}{\sin(i')}$$

$$\sin(i') = \frac{n}{n'} \sin(i)$$

$$\cos(i) = \sqrt{1 - \sin^2(i)}$$

$$\cos(i') = \sqrt{1 - \left(\frac{n}{n'}\right)^2 \sin^2(i)}$$

When  $i$  varies,  $SA'$  is not constant: the refracted rays do not pass through the same point A'. In this case, the image is blurred, and the plane diopter is not stigmatic.

## II. A Plane Diopter under Gaussian Conditions:

Incident rays at small angles ( $i \rightarrow 0^\circ$ ,  $\sin(i) \rightarrow 0$ ).

$$SA' = SA \frac{n'}{n} \sqrt{\frac{1 - \left(\frac{n}{n'}\right)^2 \sin^2(i)}{1 - \sin^2(i)}} \rightarrow \frac{\overline{SA'}}{\overline{SA}} = \frac{n'}{n}$$

The resulting relationship, called the **conjugation relation**, shows that  $\overline{SA}$  and  $\overline{SA'}$  always have the same sign and, consequently, that the object A and its image A' are in the same medium and are always of opposite nature.

## V. Plane-Parallel Plate

### I. Definition

A plane-parallel plate consists of a transparent, homogeneous medium bounded by two flat, parallel surfaces. Each of its faces may be placed either in the same medium or in different media. In the general case, where the surrounding media have different refractive indices ( $n_1$  and  $n_3$ ), an incident ray striking the plate at an angle of incidence  $i_1$  is refracted once at the entrance face ( $i_2$ ) and then a second time at the exit face, emerging at an angle  $i_3$ .

**Entrance face:**  $n_1 \cdot \sin(i_1) = n_2 \cdot \sin(i_2)$



**Exit face:**  $n_2 \cdot \sin(i_2) = n_3 \cdot \sin(i_3)$

**Exit face:**  $n_2 \cdot \sin(i_2) = n_3 \cdot \sin(i_3)$

Since the angles of incidence and refraction inside the plate are equal.

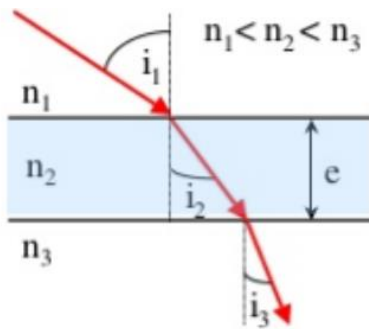


Fig.V.1.

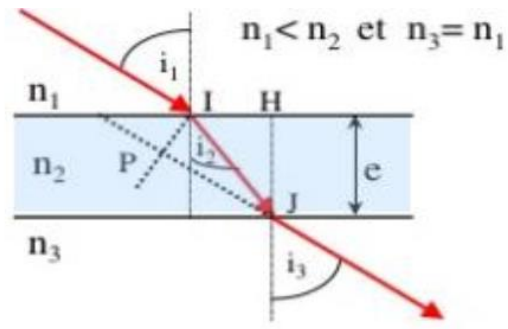


Fig.V.2.

The most interesting case is when the surrounding media are the same ( $n_1 = n_3$ ).

$$n_1 \cdot \sin(i_1) = n_3 \cdot \sin(i_3) \Rightarrow i_1 = i_3,$$

The emergent ray is then parallel to the incident ray: there is no deviation. ( $D = i_3 - i_1 = 0^\circ$ ), The ray undergoes only a lateral displacement IP.

### II. Calculation of the lateral displacement:

We assume that  $n_1 = n_3$  ( fig.v.2.),  $IJ = \frac{e}{\cos i_2}$ ,  $\widehat{IJP} = (i_1 - i_2)$  and  $IP = IJ \sin \widehat{IJP}$

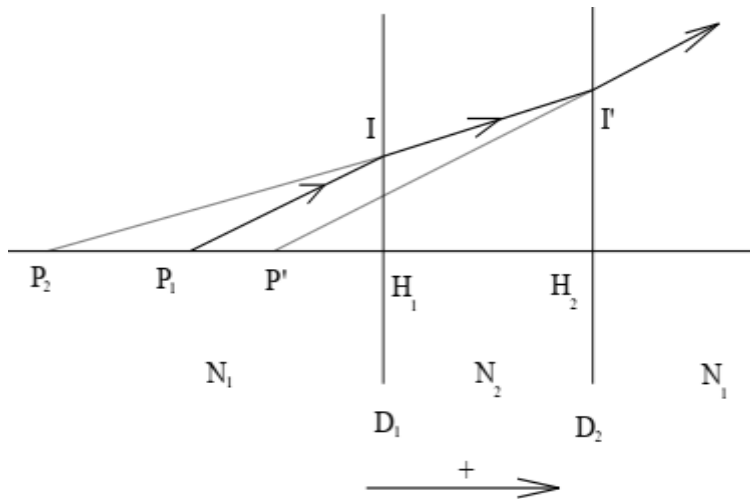
Thus :  $d = \frac{e \sin(i_1 - i_2)}{\cos i_2}$

if :  $i_1 = 0$  ;  $i_2 = 0$  and  $d = 0$  (An obvious result a priori: a ray striking at normal incidence is not deviated).

If: Si  $i_1 = \frac{\pi}{2}$   $d = \frac{e \sin(\frac{\pi}{2} - i_2)}{\cos i_2} = e$

$d$  increases with  $i_1$  and reaches a maximum for  $i_1 = \pi/2$  ; the maximum is equal to the thickness of the plate.

## II. Image of a Point



**Fig.V.3.**

As with the plane diopter, exact stigmatism occurs only for an object point at infinity. There is approximate stigmatism for central rays. We adopt this assumption and will apply the approximate formula to each of the two plane diopters that make up the plane-parallel plate  $\frac{p_1}{N_1} - \frac{p_2}{N_2} = 0$ , Taking careful account of the meaning of the different symbols in this formula. The object  $P_1$  forms, with diopter  $D_1$ , an image  $P_2$ , and we have:  $\overline{H_1 P_2} = \overline{H_1 P_1} \frac{N_2}{N_1}$ . This image  $P_2$  (Fig. V.3) acts as an object (virtual) for  $D_2$ , located in a medium of refractive index  $N_2$ . The virtual object  $P_2$  produces, through  $D_2$ , an image  $P'$  such that :

$$\overline{H_2 P'} = \overline{H_2 P_2} \frac{N_1}{N_2}$$

We now expand this relation:

$$\overline{H_2H_1} + \overline{H_1P'} = \overline{H_2H_1} \frac{N_1}{N_2} + \overline{H_1P_2} \frac{N_1}{N_2}$$

Or equivalently:

$$\overline{P_1H_1} + \overline{H_1P'} = \overline{H_2H_1} \frac{N_1}{N_2} - \overline{H_2H_1}$$

and since:  $\overline{H_2H_1} = -1$ ,  $\overline{P_1P'} = 1 \left(1 - \frac{N_1}{N_2}\right)$  which defines the position of the image relative to that of the object.



## VI. The Prism

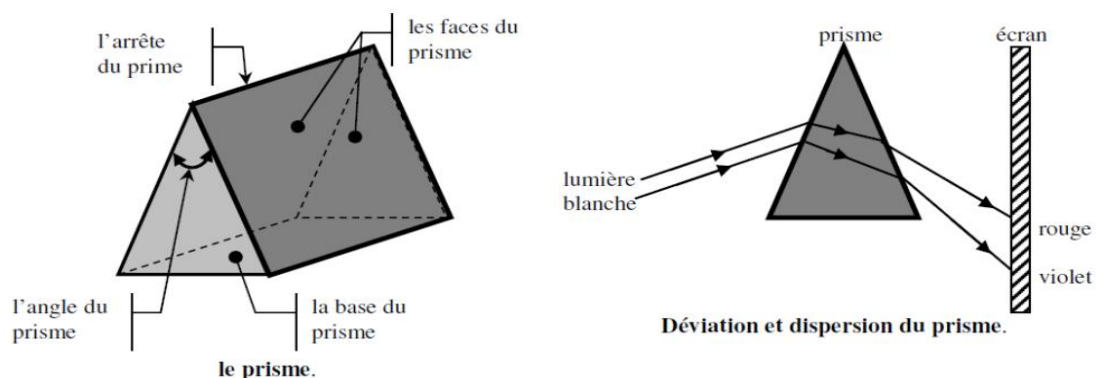
### I. Definition

A prism is an optical element used to refract, reflect, or disperse light into its components (the different wavelengths of the rainbow for white light).



**Fig. VI.1. Dispersion of light by the prism**

Prism consists of three transparent, homogeneous media separated by two non-parallel plane diopters called the faces of the prism. The third surface is called the base of the prism, which does not participate in the propagation of light. The plane diopters are the segments AB and AC, forming a triangle. The angle formed by the two faces of the prism is called the prism angle  $[A = (AB, AC)]$ . A prism has two main properties: **dispersion** and **deviation**.

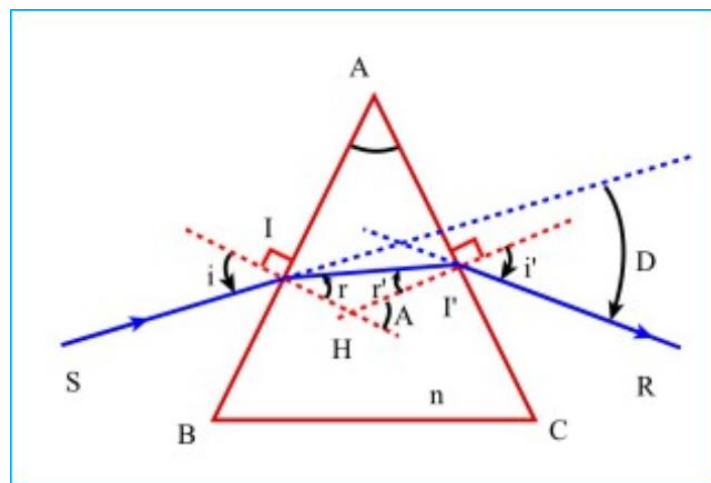


**Fig. VI.2. The Prism**

When light passes from air into glass, for example, it is refracted. When it exits through the other face, it is refracted again. The incident ray or beam is therefore deviated. However, the refractive index is not the same for different wavelengths. As a result, a beam of white light is separated into its components: blue is deviated more than yellow, which is itself deviated more than red. Under these conditions, the prism can be used to analyze polychromatic visible radiation (spectroscopy).

## II. Prism Formulas and Ray Path

Consider a prism defined in its principal section plane by its angle  $A$  and its relative refractive index  $n > 1$ . Recall that, by assumption, the surrounding medium is air. Let  $SIS$  be an arbitrary incident ray striking the entrance face  $AB$  of the prism at  $I$ , coming from a medium less refractive than the prism, and striking the exit face  $AC$  at  $I'$ . This ray undergoes refraction at both  $I$  and  $I'$  in accordance with Descartes two laws of refraction.



**Fig. VI.3. Path of a light ray through a prism**

- $A$  : Prism angle
- $i$  : Angle of incidence on the entrance face
- $r$  : Angle of refraction on the entrance face
- $r'$  : Angle of incidence on the exit face
- $i'$  : Angle of emergence on the exit face, when it exists

- D: Angle through which the extension of the incident ray must be rotated to coincide with the emergent ray (deviation angle)
- I: Point of incidence on the entrance face
- I' : Point of incidence on the exit face

According to Snell's laws at I and I', the following two relations are imposed:

At point I, we have:  $\sin(i) = n.\sin(r)$  (1)

At point I', we have:  $\sin(i') = n.\sin(r')$  (2)

In triangle IHI':  $A = r + r'$  (3)

By the definition of triangle properties, the exterior angle A is equal to the sum of the two interior angles that are not adjacent to it. The deviation angle D is, by definition, the angle through which the incident ray SI must be rotated to align with the direction of the emergent ray I'R. (Deviation is thus the angle between the extension of the incident ray and the emergent ray.) This deviation is therefore the sum of two successive deviations occurring in the same direction, one at the entrance and the other at the exit of the prism, namely:

$$D = (i - r) + (i' - r')$$

As:  $A = r + r'$

Thus, the deviation angle D is given by the formula:

$$D = i + i' - A \quad (4)$$

**A retenir**

$$\sin(i) = n.\sin(r)$$

$$\sin(i') = n.\sin(r')$$

$$r + r' = A$$

$$D = i + i' - A$$

The four formulas mentioned are the fundamental formulas of the prism. They have just been established for a particular case of the figure.

### III. Minimum Deviation

It can be shown that  $D$  reaches a minimum  $D_m$  when:

- $i = i' = \frac{D_m + A}{2} \equiv i_m$        $D = i + i' - A$
- $r = r' = \frac{A}{2} \equiv r_m$        $A = r + r'$
- $\sin i_m = n \sin r_m$        $\rightarrow$        $n = \frac{\sin\left(\frac{D_m + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

The refractive index of the prism can be determined, for a given monochromatic radiation, by measuring the minimum deviation angle.

## VII. Spherical Diopter



## I. Definitions

A spherical diopter is a portion of a refracting spherical surface separating two homogeneous, transparent media with different refractive indices. It is characterized by:

- The center C of the sphere, called the **dioptr center**
- The point S, called the **vertex of the diopter**
- The **optical axis**, the axis of rotational symmetry of the diopter, passing through points C and S.
- The radius of the sphere  $R = \overline{SC}$ , called the **radius of curvature**, an algebraic quantity that is negative for a concave spherical diopter ( $\overline{SC} < 0$ ) and positive for a convex spherical diopter ( $\overline{SC} > 0$ )

The optical axis is oriented in the direction of light propagation (from left to right).

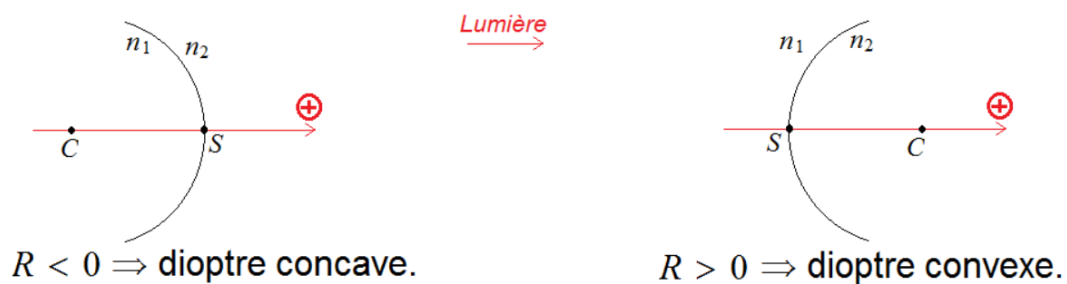


Fig.VII.1.

## II. Converging or Diverging Diopter

In a **converging spherical diopter**, the ray, after passing through the diopter, moves closer to the optical axis. Conversely, in a **diverging diopter**, it moves away from the axis.

If a **convex diopter** separates a second medium more refractive than the first ( $n_1 < n_2$ ), it is converging; otherwise ( $n_1 > n_2$ ), it is diverging.

If a **concave diopter** separates a second medium more refractive than the first ( $n_1 < n_2$ ), it is diverging; otherwise ( $n_1 > n_2$ ), it is converging.

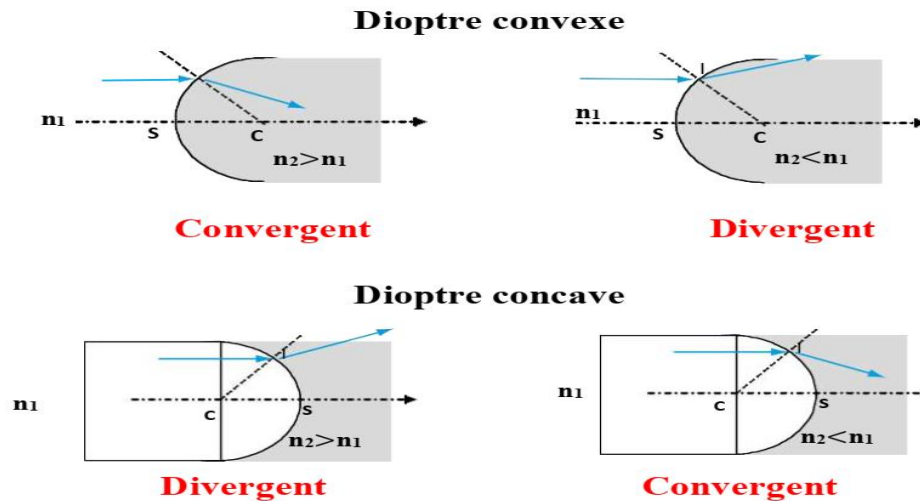
Four possible cases for the spherical diopter:

$\overline{SC} < 0$  and  $n_1 < n_2 \rightarrow$  **Concave diverging spherical diopter**

$\overline{SC} < 0$  and  $n_1 > n_2 \rightarrow$  **Concave converging spherical diopter**

$\overline{SC} > 0$  and  $n_1 < n_2 \rightarrow$  **Convex converging spherical diopter**

$\overline{SC} > 0$  and  $n_1 > n_2 \rightarrow$  **Convex diverging spherical diopter**



**Fig. VII.2. The four possible configurations of a spherical diopter**

## II. Vergence

Vergence is a parameter that characterizes the focusing properties of a centered optical system. It is an algebraic quantity, homogeneous to the inverse of a length, and it is expressed in diopters ( $\delta$ ) or in  $\text{m}^{-1}$  (1diopter= $1\text{m}^{-1}$ ). For spherical diopters, the vergence is given by the relation:

$$V = \frac{n_2 - n_1}{\overline{SC}}$$

If  $V > 0$ , the system is **converging**. A ray incident parallel to the optical axis emerges while approaching the axis, provided that it emerges on the same side of the optical axis as the incident ray.

If  $V < 0$ , the system is **diverging**. A ray incident parallel to the optical axis emerges while moving away from the axis, provided that it emerges on the same side of the optical axis as the incident ray.

Finally, if  $\mathbf{V} = \mathbf{0}$ , the system is **afocal**. A ray incident parallel to the optical axis always emerges parallel to the axis.

#### IV. Approximate stigmatism

- The spherical diopter is strictly stigmatic only for points on its surface and at its center.
- There is approximate stigmatism for any point in space that sends onto the spherical diopter only a light pencil whose mean ray is normal to it.

#### V. Conjugation relations

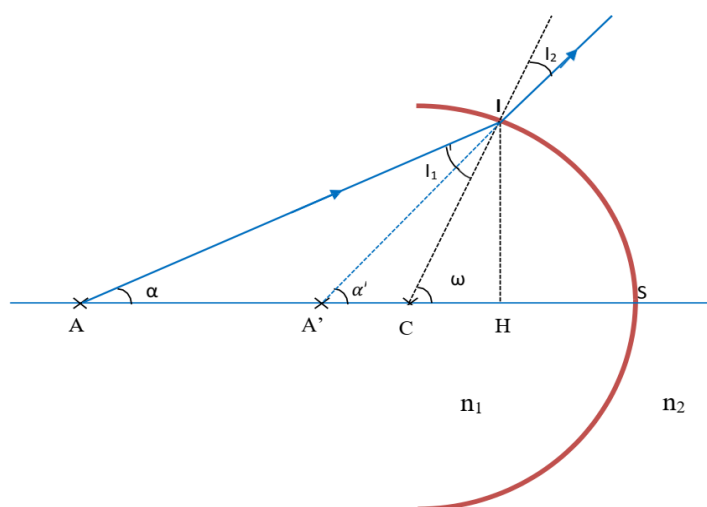
A spherical diopter is not strictly stigmatic. However, within the framework of the **Gauss approximations**, one can consider that all rays originating from an object point  $A$  intersect, after passing through the diopter, at a single point  $A'$ .

We will use this property to determine a **conjugation relation**.

**Note:** Under Gauss conditions (rays making small angles with the axis and entering the system close to the axis).

##### V.1. Conjugation relation with origin at the vertex

Let us consider a real object point  $A$  located on the optical axis of a concave diopter. The image  $A'$  of  $A$  is located at the point of intersection of two arbitrary light rays originating from  $A$ . Consider the ray emitted from  $A$  that is refracted at point  $I$  in accordance with the laws of refraction.  $A'$  is located at the point of intersection between the extension of the refracted ray and the optical axis.



**Fig. VII.3. Conjugation relation with origin at the vertex**



The normal to the diopter at point **I** is **CI**. The ray **AI** reaches the diopter with an angle of incidence **i<sub>1</sub>**. It is refracted with an angle **i<sub>2</sub>** such that **n<sub>1</sub> sin i<sub>1</sub> = n<sub>2</sub> sin i<sub>2</sub>**.

This ray intersects the optical axis at **A'**. The segment **IA'** is virtual.

Within the **Gauss approximation**, the angles **i<sub>1</sub>**, **i<sub>2</sub>**, **α**, and **ω** are sufficiently small, and points **H** and **S** can be considered coincident.

In triangles **AIC** and **A'IC**, the sum of the interior angles must be equal to **π**, that is:

$$i_1 + \alpha + (\pi - \omega) = \pi$$

$$\text{And therefore: } i_1 = \omega - \alpha,$$

$$i_2 + \alpha' + (\pi - \omega) = \pi$$

$$\text{And therefore: } i_2 = \omega - \alpha',$$

According to **Snell's law** and the **Gauss condition**, we have:

$$n_1 i_1 = n_2 i_2$$

$$\text{And therefore: } n_1 (\omega - \alpha) = n_2 (\omega - \alpha')$$

We also have:

$$\alpha' = \tan \alpha' = \frac{\overline{SI}}{\overline{SA'}}$$

$$\omega = \tan \omega = \frac{\overline{SI}}{\overline{SC}}$$

We can then find the following expression:

$$\frac{\overline{SI}}{\overline{SA'}} - \frac{\overline{SI}}{\overline{SA}} = \frac{n_2 - n_1}{\overline{SC}} = V$$

$\overline{SA'}$  ,  $\overline{SA}$  et  $\overline{SC}$  : The distances are **algebraic**, measured from **S**. This relation is called the **conjugation relation with origin at the vertex**.

## V.2. Conjugation relation with origin at the center

Starting from **Descartes' conjugation relation** with the origin at the vertex, we first obtain:

$$\frac{n_2}{\overline{SA'}} - \frac{n_1}{\overline{SA}} = \frac{n_2 - n_1}{\overline{SC}}$$

Then, by applying **Charles' relations**, we get:

$$\overline{SA} = \overline{SC} + \overline{CA} \quad \text{et} \quad \overline{SA'} = \overline{SC} + \overline{CA'} ; \text{ we can show that:}$$

$$\frac{n_2}{\overline{CA}} - \frac{n_1}{\overline{CA'}} = \frac{n_2 - n_1}{\overline{CS}}$$

This relation is called the **conjugation relation with origin at the center**.

## VI. Foci of the spherical diopter

A spherical diopter is characterized by two particular points called **foci**:

**VI.1. The object focus** represents the position of the object when its image is formed at infinity.

**VI.2. The object focal length** is then given by:

$$\overline{SF} = \frac{n_1}{n_1 - n_2} \overline{SC}$$

**VI.3. The image focus** represents the position of the image when the object is placed at infinity.

**VI.4. The image focal length** is then given by:

$$\overline{SF'} = \frac{n_2}{n_2 - n_1} \overline{SC}$$

- Any incident ray originating from **F** emerges **parallel to the optical axis**.
- Any incident ray **parallel to the optical axis** emerges **converging toward F'**.

In a spherical diopter, the **object and image focal lengths** are defined as the following **algebraic quantities**:

$$f = \overline{SF} = -\frac{n_1}{V}$$
$$f' = \overline{SF'} = \frac{n_2}{V}$$

Where **n<sub>1</sub>** and **n<sub>2</sub>** are the refractive indices of the media before and after the system. If the two media are identical, the focal lengths are **opposite in sign**.

- If **V > 0**, then **f' > 0** and **f < 0**.
- If **V < 0**, then **f' < 0** and **f > 0**.

In practice, the **image focal length f'** is mainly used to characterize the system.

✓ Object **A** at optical infinity → image at the **principal image focus F'**

$$\frac{1}{\overline{SA}} \rightarrow 0, \quad \frac{n_2}{\overline{SA'}} - 0 = \frac{n_2}{\overline{SF'}} = \frac{n_2 - n_1}{\overline{SC}} = V$$

**The image focal length**

$$f' = \overline{SF'} = \frac{n_2}{n_2 - n_1} \overline{SC}$$

✓ Image **A'** at optical infinity → object at the **principal object focus F**

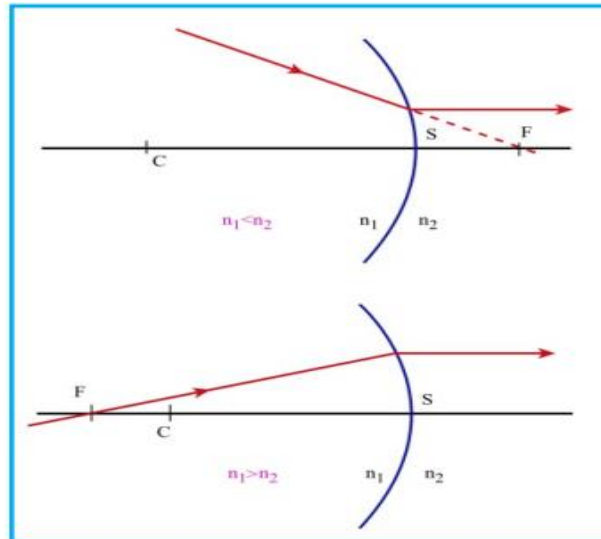
$$\frac{1}{\overline{SA'}} \rightarrow 0, \quad 0 - \frac{n_1}{\overline{SA}} = -\frac{n_1}{\overline{SF}} = \frac{n_2 - n_1}{\overline{SC}} = V$$

**The object focal length**

$$f = \overline{SF} = \frac{n_1}{n_1 - n_2} \overline{SC}$$

**Note:**

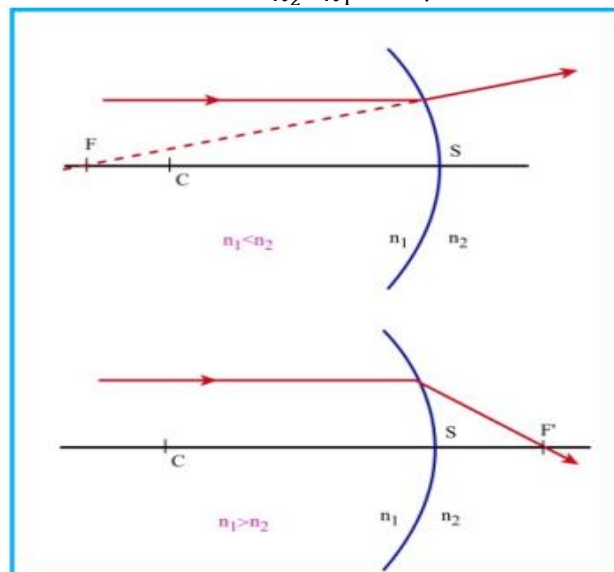
For a spherical diopter,  $f \neq f'$ .



**Fig. VII.4. Object focus F**

**Object focus F:** is the object point whose image is formed at infinity. The **object focal length f** is the **algebraic distance  $\overline{SF}$** .

$$\overline{SF} = \frac{n_1 \overline{SC}}{n_2 - n_1} = -\frac{n_1}{V}$$



**Fig. VII.5. Image focus F'**

- **Image focus F'**: is the image point of an object located at infinity. The **image focal length f'** is the **algebraic distance**  $\overline{SF'}$ .

$$\overline{SF'} = \frac{n_2 \overline{SC}}{n_2 - n_1} = \frac{n_2}{V}$$

We note that:  $\frac{\overline{SF'}}{\overline{SF}} = -\frac{n_2}{n_1} < 0$

$\overline{SF'}$  and  $\overline{SF}$  have **opposite signs**; **F** and **F'** belong to **two different media**.

## VII. Magnification

It is the **ratio of the image size to the object size**:

$$\gamma = \frac{\overline{A'B'}}{\overline{AB}}$$

- **Origin at the vertex**

$$\gamma = \frac{n}{n'} \frac{\overline{SA'}}{\overline{SA}}$$

- **Origin at the center**

$$\gamma = \frac{\overline{CA'}}{\overline{CA}}$$

- **Origin at the foci**

$$\gamma = -\frac{\overline{SF}}{\overline{FA}} = -\frac{\overline{S'A'}}{\overline{SF'}}$$

- If  $\gamma > 0$  (+), the image is **upright** (same orientation as the object).
- If  $\gamma < 0$  (−), the image is **inverted** (opposite orientation).
- If  $|\gamma| > 1$ , the image is **larger** than the object.
- If  $|\gamma| < 1$ , the image is **smaller** than the object.
- If  $\gamma = 1$ , the image and the object are of the **same size**.

## VII. Characteristics of the image

### VII.1. Image position $\overline{SA'}$

- If  $\overline{SA'} > 0$ , the image is **real**.
- If  $\overline{SA'} < 0$ , the image is **virtual**.

### VII.2. Nature of the image

- **Real or virtual**: Determine whether the image is formed on the same side as the outgoing rays (real) or on the side of the incident rays (virtual).
- **Upright or inverted**: Indicate if the image has the same orientation as the object (upright) or opposite (inverted).
- **Magnification**: Specify whether the image is **enlarged**, **reduced**, or of the **same size** as the object.

## VIII. Image Construction

Geometric construction is essential to **visualize and verify** the results obtained by calculation.

We always consider an object  $A_1B_1$  perpendicular to the optical axis; its associated image will be  $A_2B_2$ . To minimize errors, it is preferable to draw the following **three special rays**:

- The ray from  $B_1$  parallel to the optical axis is refracted through the **image focus  $F'$** .
- The ray from  $B_1$  passing through the **object focus  $F$**  emerges **parallel to the optical axis**.
- The ray from  $B_1$  passing through the **center  $C$**  of the diopter is **undeviated** ( $i = 0$  and  $r = 0$ ).

### VIII.1. Case of a converging diopter (real object)

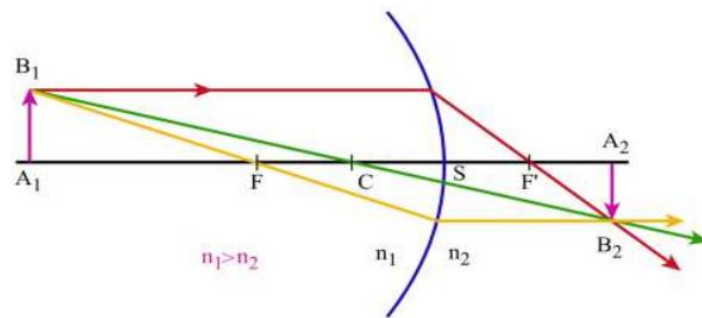


Fig. VII.6. Geometric construction of the image  $A_2B_2$  for a converging spherical diopter

### VIII.2. Case of a diverging diopter (real object)

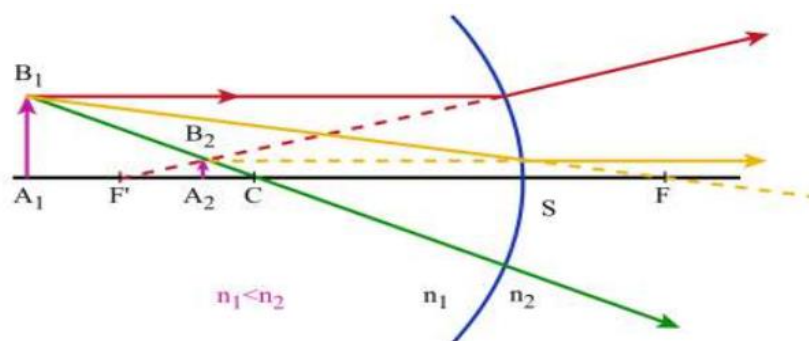


Fig. VII.7. Geometric construction of the image  $A_2B_2$  for a diverging spherical diopter

## VIII. Thin Lenses



## Thin Lenses

Lenses are essential components of optical systems. They have made it possible to explore the infinitely large (telescopes) and the infinitely small (microscopes), as well as to correct vision (glasses).

They are found both in everyday life (eyeglasses, contact lenses, cameras) and in scientific research (telescopes, spectrographs, microscopes, etc.).

### Description

A **thin lens** is formed by the combination of **two spherical diopters**, or a **spherical diopter and a plane diopter**, with **large radii of curvature ( $R_1, R_2$ )** relative to the **lens thickness ( $e$ )**.

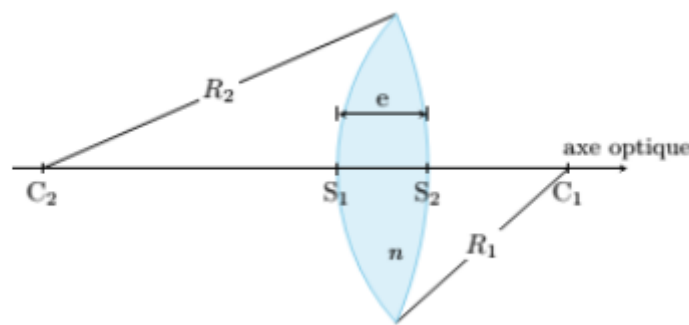
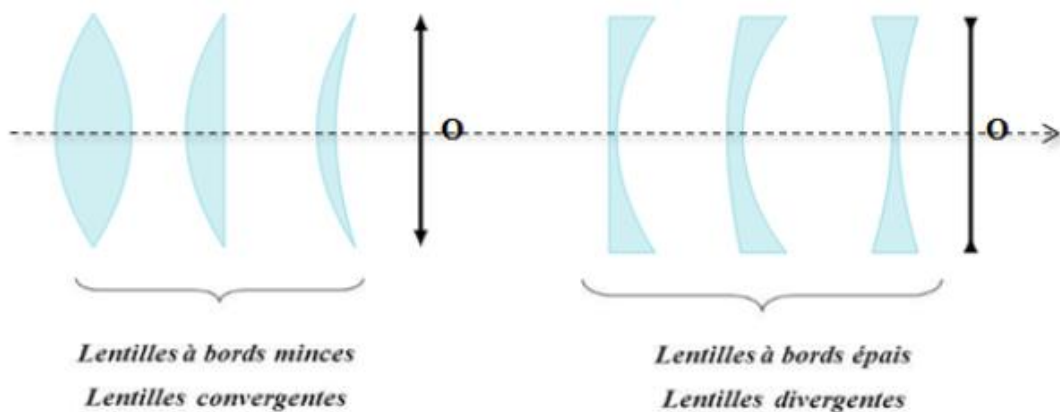


Fig. VIII.1. Optical lens

In the **thin lens approximation**, the vertices  $S_1$  and  $S_2$  are considered coincident at a point **O**, called the **optical center**.

There are two types of lenses:

- **Thin-edged lenses**, which are **converging**.
- **Thick-edged lenses**, which are **diverging**.



## Characteristics of Thin Lenses

### Optical Center

The **optical center** of the lens, denoted **O**, is defined as the point on the optical axis of the lens through which passes the refracted ray corresponding to an incident ray whose emergent ray is parallel to it.

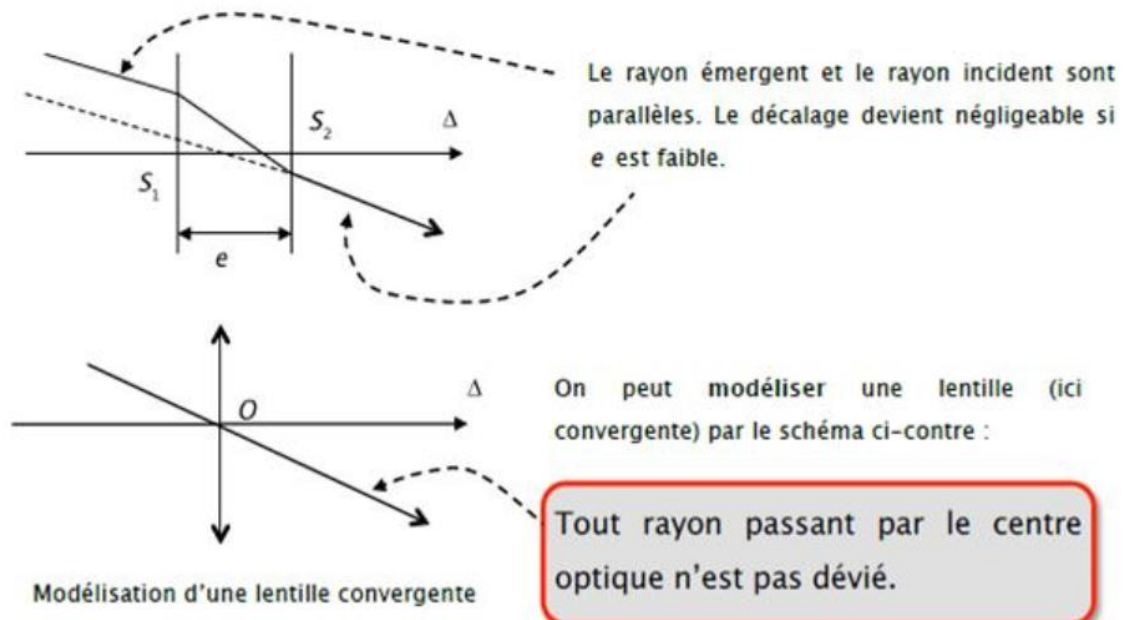


Fig. III.8 Ray passing through the center O.

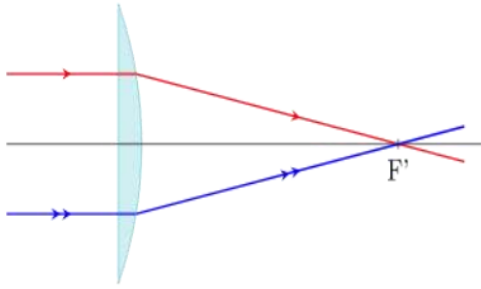
### Foci

#### Case of a converging (thin) lens

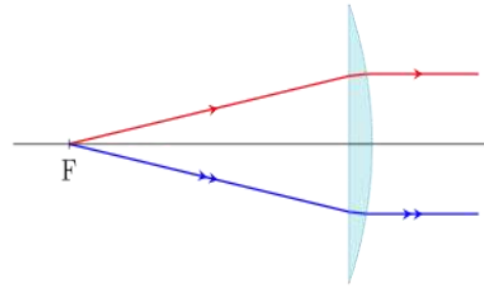
A converging lens has two foci, called the **principal object focus** and the **principal image focus**:

- Any incident ray passing through **F**, the principal object focus, emerges **parallel to the optical axis**. Thus, this focus corresponds to an object whose image is formed at infinity.
- Any incident ray parallel to the optical axis emerges passing through **F'**, the principal image focus. This focus is therefore the image of an object located at infinity.
- These foci are **symmetrical** with respect to the optical center of the lens.





**Fig. 9.4: Principal Image Focus of a converging lens**

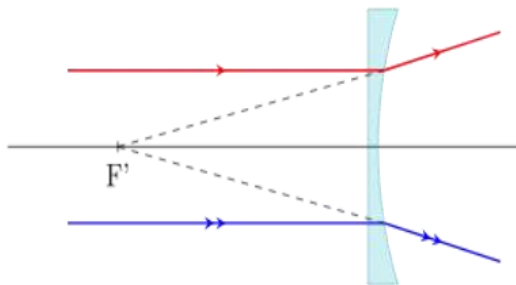


**Fig. 9.5: Principal Object Focus of a converging lens**

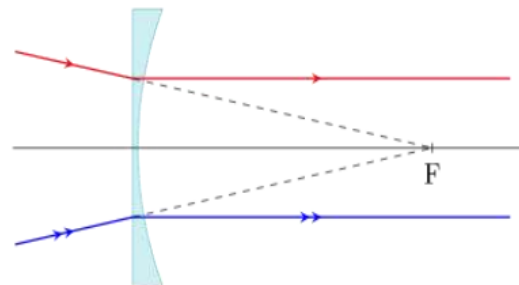
### Case of a diverging (thick) lens

A diverging lens also has two foci, whose positions are **reversed** compared to those of a converging lens:

- Any incident ray whose **extension passes through F**, the principal object focus, emerges **parallel to the optical axis**.
- Any incident ray **parallel to the optical axis** emerges so that its **extension passes through F'**.
- These foci are also **symmetrical** with respect to the optical center of the lens.



**Fig. 9.6: Principal image focus of a diverging lens**

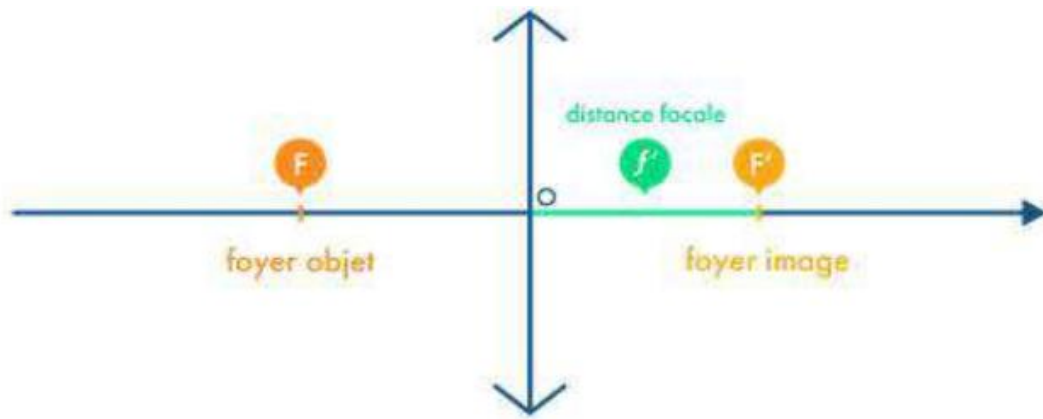


**Fig. 9.7: Principal object focus of a diverging lens**

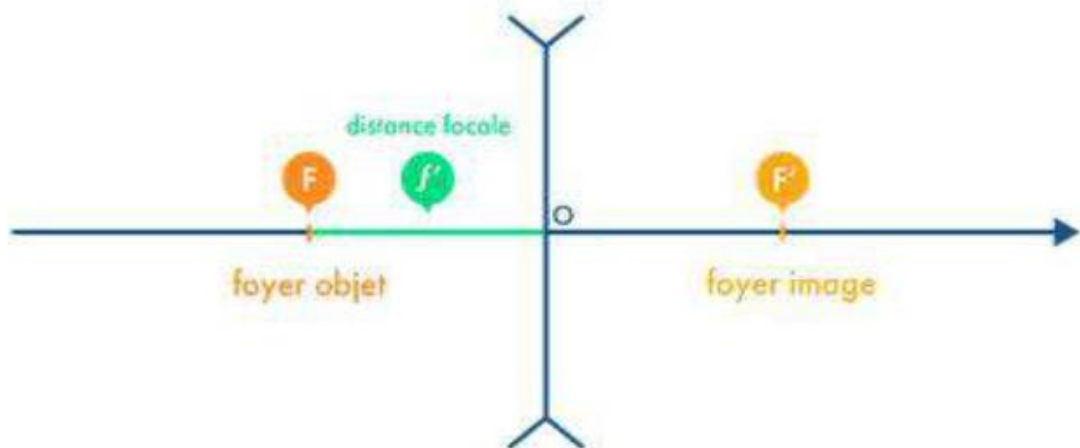
### Focal Length

The image and object focal lengths are defined as the respective algebraic quantities:

$$f' = \overline{OF'} \quad \text{and} \quad f = \overline{OF} . \quad \text{They are expressed in meters (m).}$$



**Fig. 9.8: Focal length of a converging lens**



**Fig. 9.8: Focal length of a diverging lens**

#### **Note**

- The lens is **converging** if  $f' > 0$  and  $f < 0$ .
- The lens is **diverging** if  $f' < 0$  and  $f > 0$ .

#### **9.2.4. Vergence of a Lens**

The **vergence V** of a thin lens is the **inverse of its focal length**:

$$V = \frac{1}{f'} = -\frac{1}{f} = (n-1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

It is expressed in **diopeters ( $\delta$ ) or  $\text{m}^{-1}$** . It is **positive** for a converging lens and **negative** for a diverging lens.

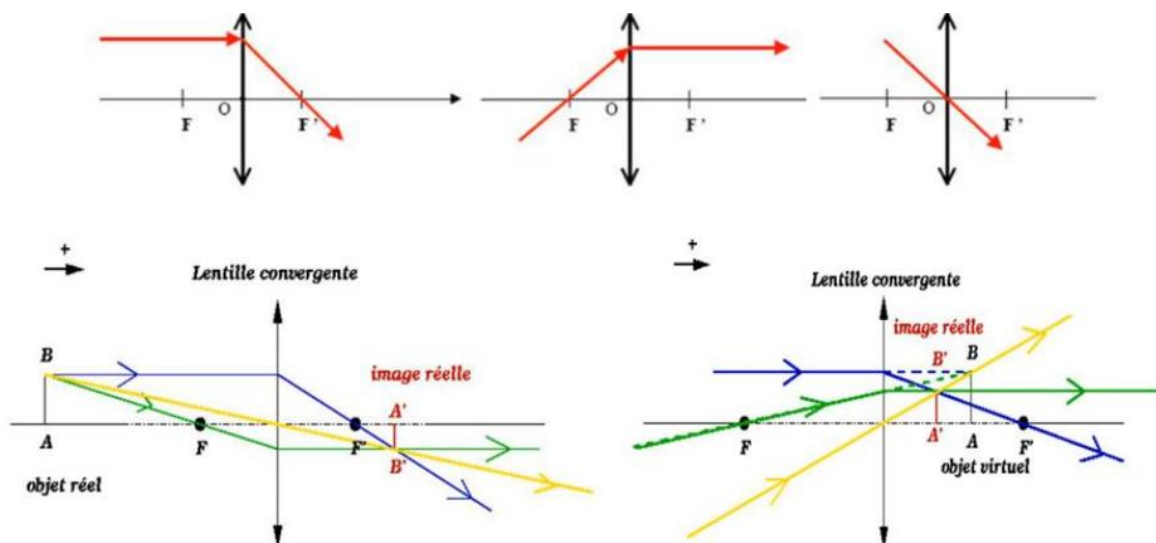
## Image Formation

### Geometric Construction of the Image

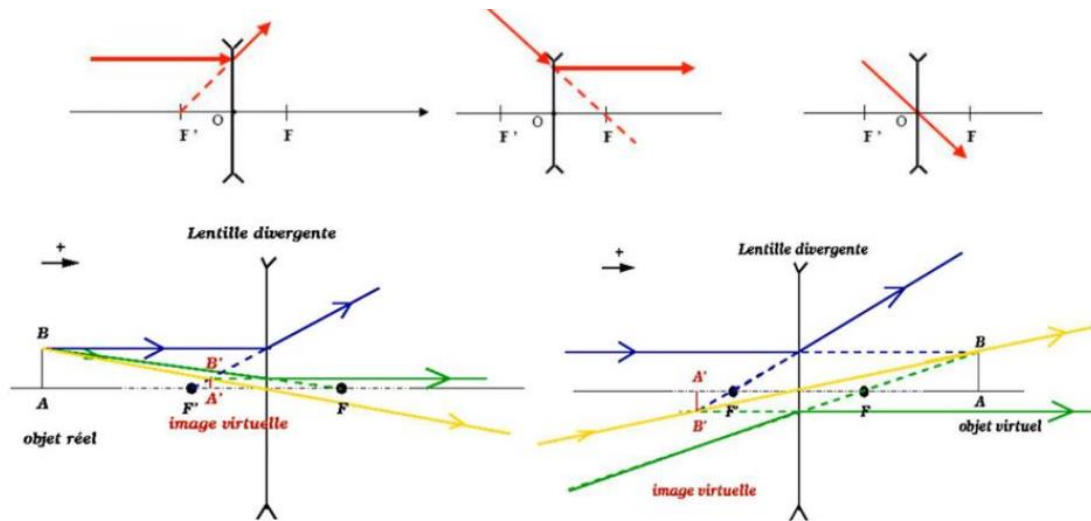
To construct the image of an extended object, the following principles are followed:

- Work within the **Gauss approximation**, meaning there is **approximate stigmatism** and **approximate aplanatism**.
- **Place the object correctly:**
  - If the object **AB** is real, it is located **to the left of the lens**.
  - If the object is **virtual**, it is located **to the right of the lens**.
- To find the image of a point, choose rays whose behavior is known:
  1. A **horizontal ray** hitting a converging lens passes through  **$F'$** ; for a diverging lens, it appears to **originate from  $F'$** .
  2. A ray passing through (or extending through)  **$F$**  emerges **horizontal**.
  3. A ray passing through the **optical center  $O$**  is **undeviated**.
- Once the rays are drawn, determine whether the image is **real or virtual**:
  - If the rays from **B** actually intersect at  **$B'$** , the image is **real**.
  - If the rays from **B** diverge after refraction and appear to come from  **$B'$** , the image is **virtual** (visible to the naked eye).

### a) Image Construction for a Converging Lens



## b) Image Construction for a Diverging Lens



### Conjugation Formula

The conjugation formula is the relation between the position of the object ( $OA$ ) and the position of the image ( $OA'$ ).

$$\frac{1}{\overline{OA'}} - \frac{1}{\overline{OA}} = \frac{1}{f'} = -\frac{1}{f}$$

$\overline{OA'}$  : Position of image  $A'$

$\overline{OA}$  : position of the object  $A$

$f'$ : image focal length

$f$ : object focal length

### Magnification

It is the ratio of the image size to the object size.

$$\Upsilon = \frac{\overline{A'B'}}{\overline{AB}} = \frac{\overline{OA'}}{\overline{OA}}$$

$\overline{A'B'}$  : Image size

$\overline{AB}$ : object size

**Object–image size**

**Si  $\gamma > 0$  (+)** The image is upright (it has the same orientation as the object)

**Si  $\gamma < 0$  (-)** The image is inverted ( opposite in orientation)

**Si  $|\gamma| > 1$**  The image is larger than the object

**Si  $|\gamma| < 1$**  The image is smaller than the object

**Si  $\gamma = 1$**  The image and the object are the same size

## **THE HUMAN EYE AND VISION**



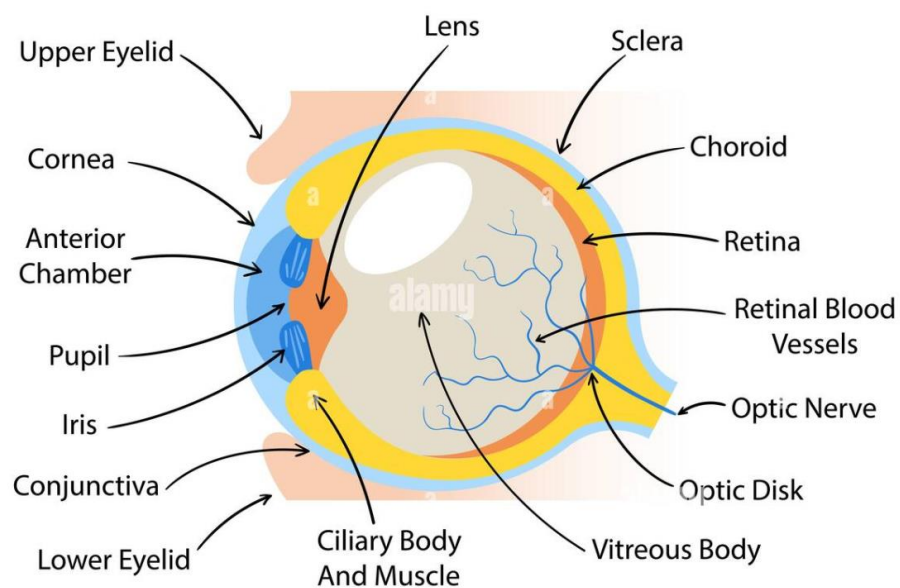
## I. The Eye: The Organ of Vision

The eye is the main organ of the visual system, which captures images and converts them into electrical signals sent to the optic nerve. These signals are then “interpreted” by the brain, at the level of the visual cortex, which reconstructs the processed image and allows us to understand our environment.

Human vision represents the brain’s primary source of information about the outside world: approximately 70% of information is received through vision. The human visual field covers about 220° horizontally and 140° vertically.

## II. Anatomy of the Eye:

The eye is a globe about 25 mm in diameter, weighing approximately 8 grams. It is composed of a multitude of structures:



**The sclera:** It is a white, opaque, and very tough membrane. It forms the “white” of the eye and is bordered at the front of the eye by the cornea.

**The cornea:** It is a transparent tissue located at the front part of the eye. It has the shape of a spherical lens (convex-concave) that transmits light to the lens. It is avascular (which explains why it does not bleed) but highly innervated.

**The choroid:** is a tissue of the eyeball, richly vascularized, that lines the inside of the sclera. It is the nourishing membrane of the eye. At the front of the eye, it transforms into the ciliary muscle, which itself continues forward to form the iris and the suspensory ligaments of the lens.

**The ciliary body:** is the anterior part of the choroid, to which the lens is attached by a series of ligaments. It plays a fundamental role in the secretion of aqueous humor and in visual accommodation.

**The lens:** is a non-homogeneous, elastic lens with variable thickness and a diameter of 10 mm, composed of layers. Its refractive index ranges from 1.35 at the edges to 1.42 along the axis. Its elasticity is essential for the mechanism of vision.

**The iris:** is a disc-shaped membrane with a central opening. It forms the colored part of the eye. The hole in its center, called the pupil, allows light to pass through to the lens. The diameter of the pupil varies according to the intensity of light (2–8 mm).

**The aqueous humor:** is a transparent fluid with a refractive index of  $n = 1.336$ . It fills the anterior chamber of the eye (the space between the cornea and the lens). It provides nutrients to the cornea and the lens and also helps maintain intraocular pressure and the shape of the eyeball.

**The vitreous humor (or body):** is a transparent, gelatinous substance with a refractive index of 1.336 that fills the posterior chamber of the eye.

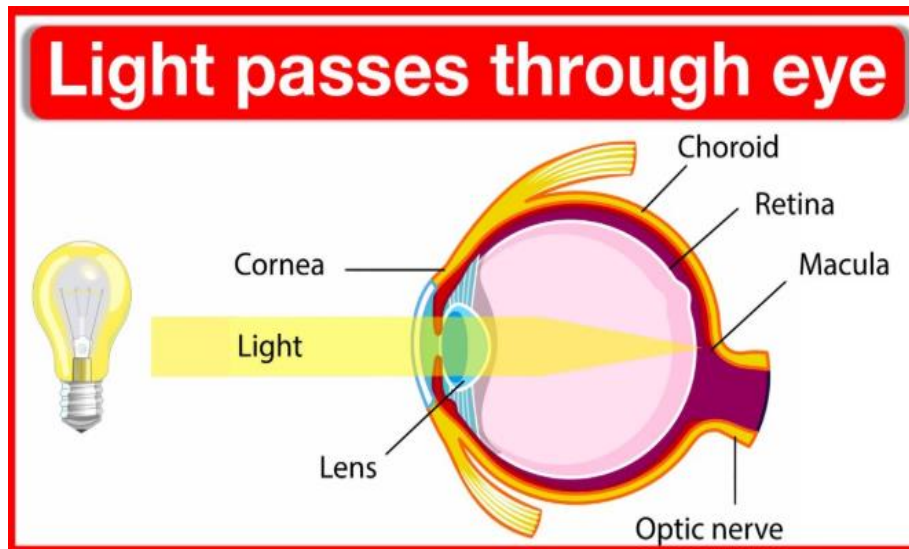
**The retina:** is a thin membrane that covers a large part of the inner surface of the eyeball. Sensitive to light, it is composed of photosensitive cells (cones and rods) and neurons that transmit electrical signals to the brain. The central retina contains the macula and the fovea. It is vascularized by the central retinal artery and vein.

- **Cones**, mainly located in the central retina (macula), are responsible for color vision, shape and detail perception, and are associated with daytime vision.
- **Rods**, mainly located in the peripheral retina, are much more sensitive to light, responsible for detecting edges and motion, and are associated with low-light (night) vision.

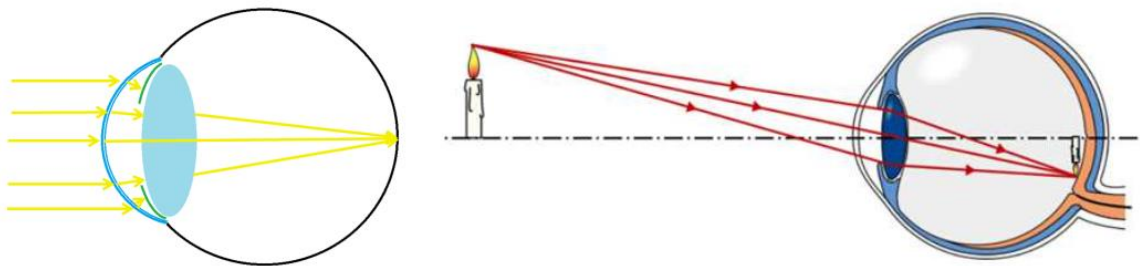
**The optic nerve** begins at the optic disc and transmits visual information from the retina to the brain.

From an optical point of view, the parts of the eye involved in image formation are: the cornea, the lens, the aqueous humor, the vitreous humor, and the retina.





### III. Path of Light in the Eye and Image Formation



Light rays first pass through the cornea, which converges them for the first time. Next, the rays pass through the aqueous humor. Some of these rays are blocked by the iris, while the rest pass through the pupil and the lens. The lens converges the rays for the final time. The rays then pass through the vitreous humor and finally reach the retina.

For the image to be sharp, it must be formed on the retina; otherwise, it will be blurry. In other words, for a point on an object, all rays originating from that point must converge at a single point on the retina for the image to be clear. If the rays converge before or after the retina, a blurred spot will form, and the image will be out of focus.

- If the eye is too long, images form in front of the retina: this is **myopia**.
- If the eye is too short, images form behind the retina: this is **hyperopia**.

- Finally, for near vision, the eye accommodates; that is, a small muscle located behind the iris, the **ciliary muscle**, contracts and changes the curvature of the lens, increasing its power so that images continue to form on the retina. Between the ages of 40 and 50, this “automatic focusing” system gradually stops working and must be supplemented with **converging lenses**: this is **presbyopia**.

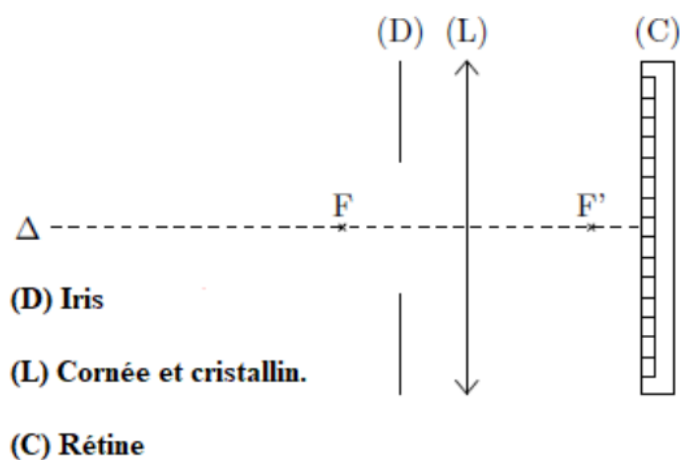
As described above, the structure of the eye is relatively complex. To simplify its optical study, we will represent it using the **reduced eye** model.

#### IV. Reduced Models of the Eye

The eye is an optical system composed of several elements. It can be considered as being made up of four spherical diopters (air–cornea, cornea–aqueous humor, aqueous humor–lens, and lens–vitreous humor) or as two lenses (cornea and lens) surrounded by different media (air, aqueous humor, and vitreous humor). This makes the study of the eye from an optical point of view quite complex. To simplify things, the eye can be replaced by **equivalent optical systems** that are much simpler. These equivalent optical systems are the **reduced spherical diopter model** and the **simplified lens model**.

#### V. Reduction to a Thin Lens

Since the eye is a converging system, it can be modeled as a **thin converging lens** placed in air (representing the cornea and lens), with a variable focal length, and a screen.



**Fig. Modeling of the human eye**

## VI. Functioning of the eye

### VI.1. Formation of an image on the retina

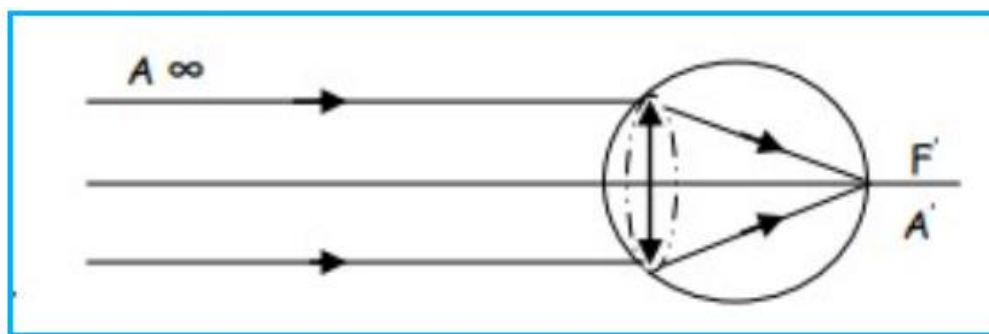
For an object to be seen clearly, its image must be formed on the yellow spot (macula) of the retina, centered on the optical axis.

**Two conditions must be fulfilled:**

- The object must not be too far from the optical axis of the eye.
- The image of the object must be formed at a constant distance (equal to 17 mm) from the optical center O.

### VI.2. Vision at infinity

In the normal eye (called emmetropic) at rest, the image focal plane coincides with the retina. The image of an object at infinity ( $\infty$ ) is therefore formed exactly on the retina, and the object is seen clearly.

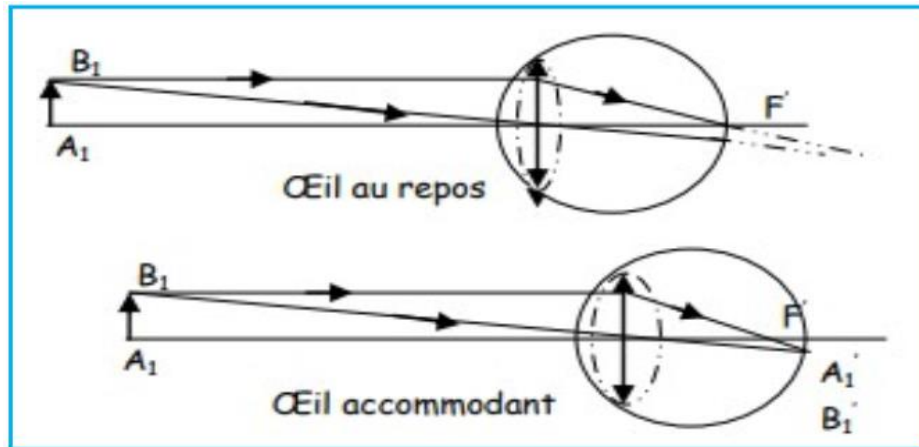


**Fig. The image of an object at infinity in the normal eye at rest**

### VI.3. Near vision: accommodation

Observing a nearby object requires a delay and an effort of visual adaptation: this is called accommodation of the eye. When the object moves closer to the eye, in the absence of accommodation, its image shifts in the same direction. It therefore forms behind the retina, and vision becomes blurred.

The eye must bring the image back onto the retina by increasing its convergence. This can be achieved through deformation of the crystalline lens under the action of the ciliary muscles.



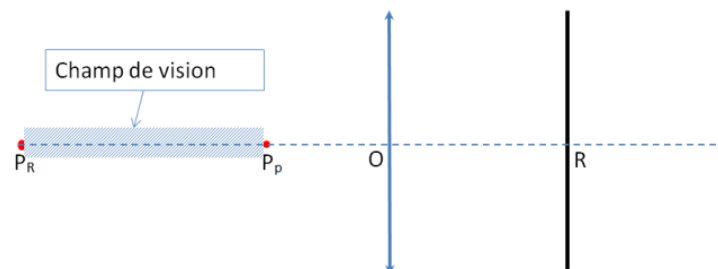
**Fig. Accommodation of the eye**

### Limits of accommodation

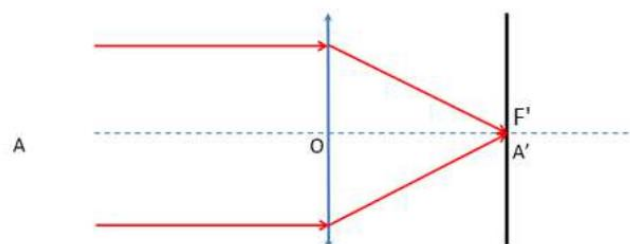
The *Punctum Remotum* (PR) of an eye is the farthest point that the eye can see clearly. The PR is real if it is located in object space; otherwise, it is virtual.

The *Punctum Proximum* (PP) is the closest point that the eye can see clearly.

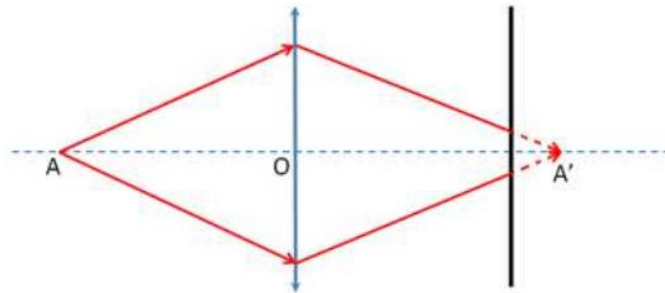
The field of vision is the interval  $[P_R, P_P]$ . It determines the different positions at which vision is clear. Outside this interval, vision is blurred.



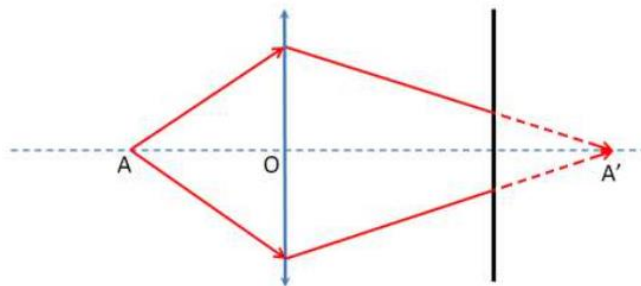
When the object is located at infinity, its image is situated at the image focus  $F'$ , and therefore on the screen.



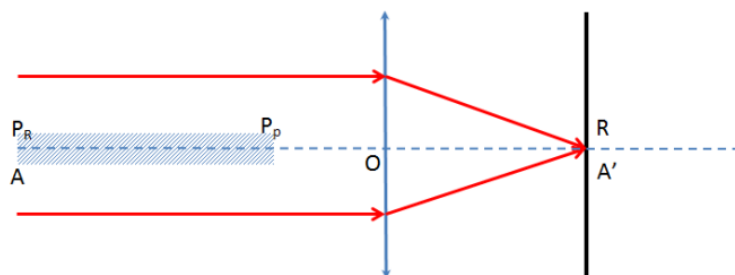
If the object moves closer to the lens, its image will shift away from its initial position and will be located behind the screen. On the screen, a blurred image of the object will be seen in the form of a small spot.



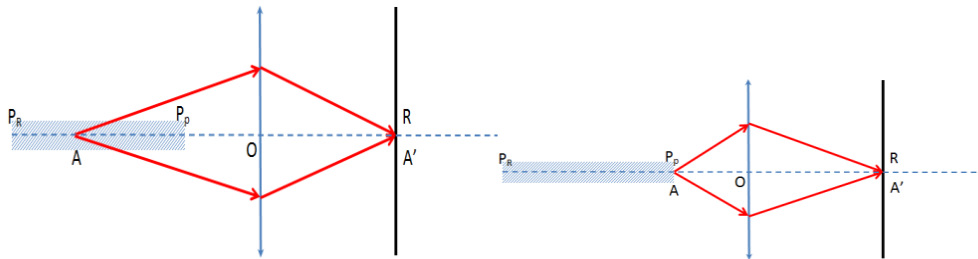
If the object moves even closer to the lens, its image will shift further from its initial position. On the screen, a more blurred image of the object will be seen, and the spot will be larger.



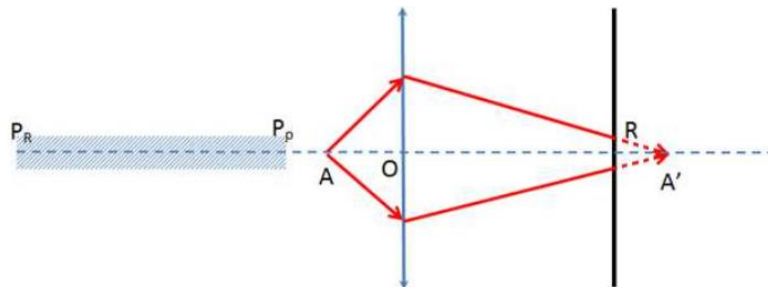
Let us now consider the eye, which we will model as a converging lens. When the object is at the PR (Punctum Remotum), its image is located on the retina.



When the object moves closer to the eye but remains within the field of vision, the image still forms on the retina.



When the object moves even closer to the eye and is located between the PP (Punctum Proximum) and the eye, its image forms behind the retina.



### In summary:

When an object is within the eye's field of vision, its image always forms on the retina, regardless of its position within this interval. When the object is outside the field of vision, its image forms either in front of or behind the retina, depending on whether it is located beyond the PR (Punctum Remotum) or between the PP (Punctum Proximum) and the eye, respectively.

To keep the image on the retina as the object moves within the field of vision, the eye changes its vergence by altering the curvature of its crystalline lens. This is achieved through pressure exerted by the ciliary muscle on the lens.

When the object is at the PR, the eye is at rest, and no pressure is applied to the lens. As the object moves closer toward the PP, the lens changes its curvature under increasing pressure from the ciliary muscle. The decrease in the lens's curvature radius increases its vergence, ensuring that the image remains on the retina. When the object is at the PP, the pressure on the lens is maximal, as is its vergence. If the object moves closer than the PP (between the PP and the eye), the lens can no longer increase its vergence, and the image forms behind the retina.

This process is called **accommodation**.

## VIII. Optical defects of the eye and their corrections

It most often involves defects in the eye's geometry or aging of the lens, which lead to blurred vision.

### VIII.I. Myopia

A myopic eye is an eye that is too long or too optically powerful, which causes light rays to converge in front of the retina. The image of a point is therefore formed as a blurred spot on the retina. In practice, this results in clear near vision and blurred distance vision.

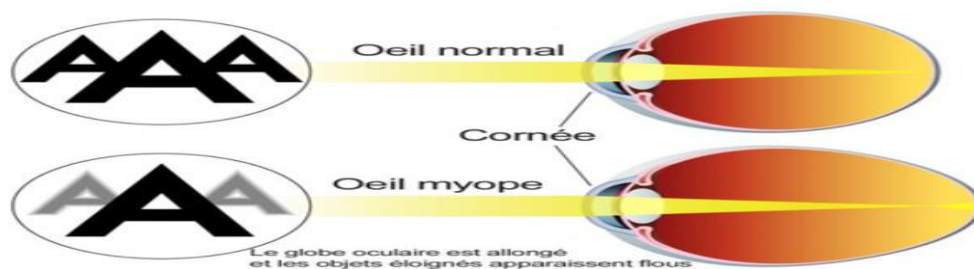


Fig. IX.7. Myopic eye

At infinity, the image is formed in front of the retina, either due to excessive convergence of the lens or the cornea, or because the eye is too long. A distant object is therefore blurred. As the object moves closer, its image moves closer to the retina. It is formed on the retina when the object reaches the **punctum remotum (PR)**, which is the farthest point that a myopic person can see clearly (the PR of a normal eye is at infinity).

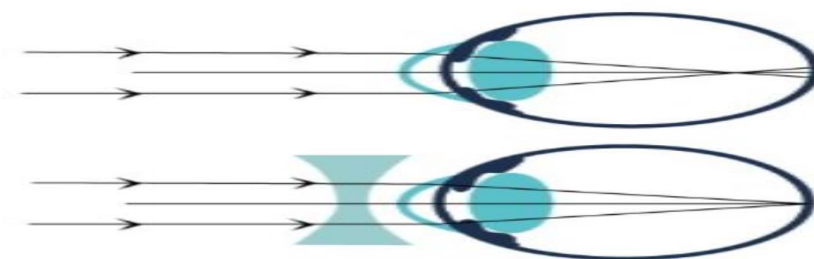
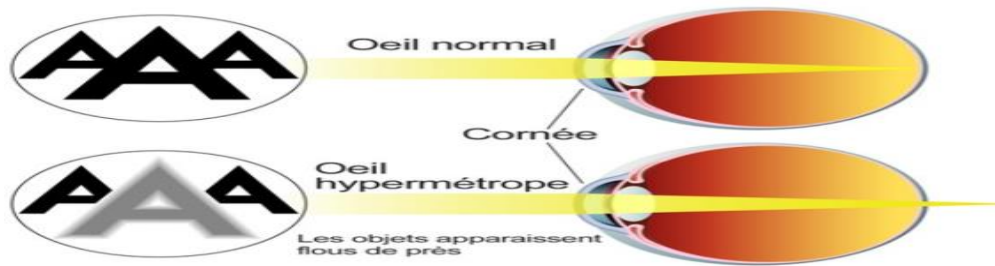


Fig. IX.8. Correction of myopia

The correction consists of placing a diverging lens in front of the eye (glasses, contact lenses) or reducing the curvature and therefore the optical power of the cornea using laser surgery.

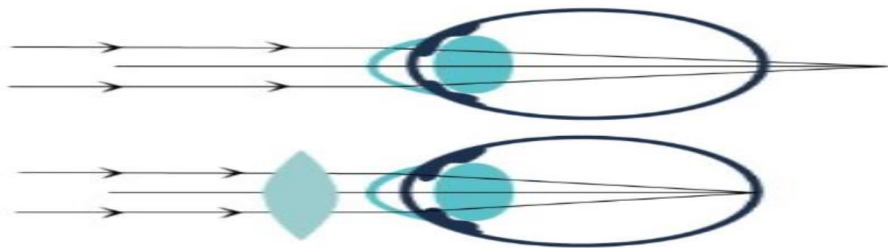
## VIII.2. Hyperopia

A hyperopic eye is an eye that is too short or not optically powerful enough, which causes light rays to converge behind the retina. The resulting image of a point is therefore a blurred spot on the retina. Accommodation allows young hyperopic individuals to compensate for this defect, resulting in clear vision at the cost of visual effort. The greater the hyperopia and the older the patient, the less this optical defect can be compensated: the patient sees neither distant nor near objects clearly.



**Fig. IX.9. Hyperopic eye**

At infinity, the image is formed behind the retina, either due to insufficient convergence of the lens or cornea, or because the eye is too short. A hyperopic eye must accommodate continuously for the image of a distant object to be clear.



**Fig. IX.10. Correction of hyperopia**

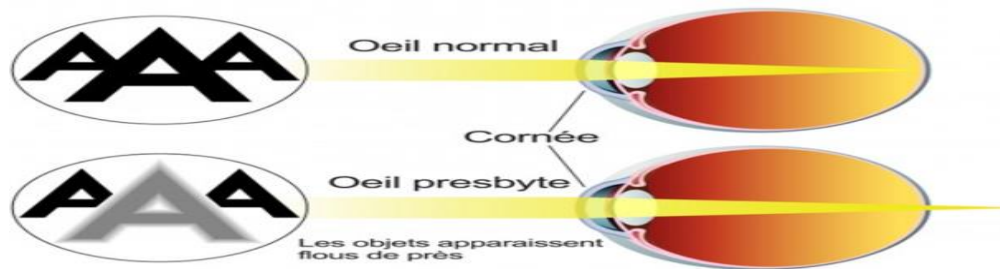
The correction consists of placing a converging lens in front of the eye (glasses) or increasing the curvature and therefore the optical power of the cornea using laser surgery.

## VIII.3. Presbyopia

With age, the eye's accommodative power decreases, meaning it can no longer focus properly for near vision. The patient must hold text progressively farther away to read. This phenomenon, which usually begins around the age of forty,

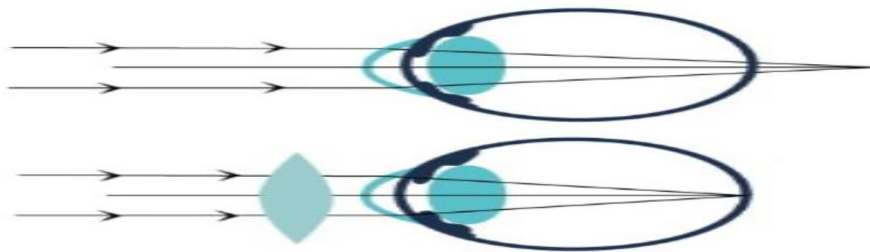


stabilizes around sixty. Laser treatment on certain areas of the cornea can be performed to create multifocality, allowing the patient to read at all distances, similar to wearing progressive lenses.



**Fig. IX.11. Presbyopic eye**

As we age, the lens loses its flexibility, causing a decrease in accommodative power. Vision of distant objects remains unchanged, but seeing nearby objects becomes difficult: the **punctum proximum** moves farther away.

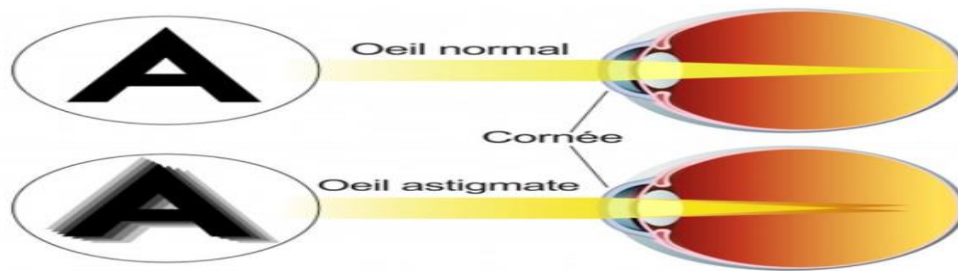


**Fig. IX.12. Correction of presbyopia**

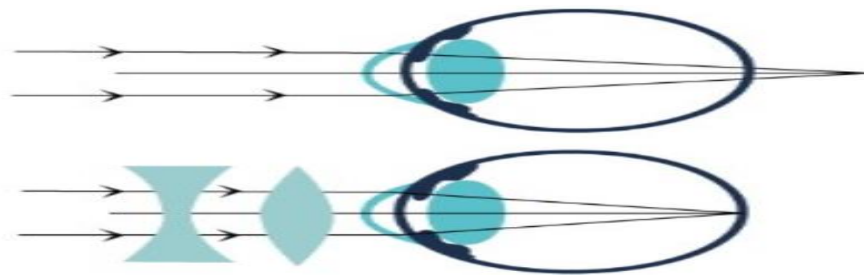
The correction consists of placing a converging lens in front of the eye for near vision. To maintain clear distance vision, **progressive lenses** can be used, which strongly correct near vision while minimally affecting distance vision.

#### **VIII.4. Astigmatism**

An astigmatic eye is an eye that is not perfectly spherical (it is more oval, like a rugby ball), which causes light rays to converge on two different planes instead of one. The resulting image of a point is therefore a distorted spot on the retina. In practice, this results in blurred vision both near and far.



**Fig. IX.13. Astigmatic eye**



**Fig. IX.14. Correction of astigmatism**

To correct astigmatism, corrective lenses or specialized contact lenses can be used, which compensate for the power along one axis to make it equal to the other.

### **Key points to remember**

A myopic eye is too convergent, a hyperopic eye is not convergent enough, a presbyopic eye accommodates less effectively, and an astigmatic eye has distorted or blurred vision at all distances.

**It is important to note that most of these vision problems can be corrected with glasses, contact lenses, or, in some cases, surgical intervention.**